Class 8

Biology

Olympiad Program
Volume - I

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1. Diversity in Living organisms

Solutions

Plants

LEVEL-I

1. Correct option: (1) a-iii, b-iv, c-ii, d-i

Chlamydomonas is a unicellular alga. Cycas is a gymnosperm, Selaginella is pteridophyte. Sphagnum is bryophyte or a moss.

2. Correct option: (B) Whitaker

Five kingdom of classification was proposed by RH Whittaker in 1969. He divided all living organisms into five kingdoms based on following features: Complexity of cell structures. Complexity of body structures.

3. Correct option: (A) Metaphyta

The Kingdom Plantae is also called as kingdom Metaphyta. The Kingdom Plantae includes all types of eukaryotic, multicellular, photosynthetic plants found in this biosphere.

4. Correct option: (D) Bacteria

Cyanobacteria are sometimes considered algae, but they are actually bacteria (prokaryotic), where the term "algae" is now reserved for eukaryotic organisms. They also derive their energy through photosynthesis, but lack a nucleus or membrane bound organelles, like chloroplasts.

5. Correct option: (A) Linnaeus

Carolus Linnaeus gave the artificial system of classification which is based on few morphological characters.

6. Correct option: (C) Both (A) and (B)

Artificial systems of classification were based upon morphological characters {such as habit, habitat, colour, number and shape of leaves, etc.) for grouping of organisms. They were mainly based on vegetative characters or on the androecium structure.

7. Correct option: (C) Both (A) and (B)

Artificial systems were the earliest systems of classification that used only gross superficial morphological characters such as habit, colour, number and shape of leaves, etc. They were based mainly on vegetative characters or on the androecium structure (system given by Linnaeus).

8. Correct option: (D) Based on mainly vegetative characters (or) on the androecium structure

Natural classification systems were based on natural affinities among the organisms and consider, not only the external features, but also internal features, like ultrastructure, anatomy, embryology and phytochemistry. Such a classification for flowering plants was followed by George Bentham and Joseph Dalton Hooker.

9. Correct option: (A) Evolutionary relationships between the various organisms

Phylogenetic classification system is based on the evolutionary relationship. As it is based on the evolution of life and the evolutionary history of a group of genetically-related organisms is called a phylogeny. A phylogeny is usually represented by phylogenetic tree.

10. Correct option: (3) a-iii, b-iv, c-ii, d-i

Artificial systems of classification were based upon morphological characters for grouping of organisms. They were mainly based on vegetative characters or on the androecium structure. Natural classification system is based on natural affinities among the organisms. It considers external features along with internal features such as ultrastructure, anatomy, embryology and phytochemistry. Phylogenetic classification is based on evolutionary relationships between organisms. Cytotaxonomy is the system of

classification of organisms which is based upon the cytological information of their cells like chromosome number.

11. Correct option: (B) Algae

Algae are chlorophyll-bearing, simple, thalloid, autotrophic and largely aquatic (both fresh water and marine) organisms.

12. Correct option: (2) i-c, ii-d, iii-a, iv-b

The science of classification is taxonomy, while the study of evolutionary connections between species is phylogeny. Numerical taxonomy is a classification system in biological systematics which deals with the grouping by numerical methods of taxonomic units based on their character states. Cytotaxonomy is the system of classification of organisms which is based upon the cytological information of their cells. These includes Chromosome number, morphology, and Behaviour.

13. Correct option: (4) a-iii, b-iv, c-i, d-ii

Spirogyra belongs to Chlorophyceae and has a filamentous structure. Colonial forms exist in case of volvox. Kelps have massive plant body reaching up to the surface of water body. Chlamydomonas are unicellular green algae.

14. Correct option: (D) All of the above

Vegetative reproduction usually takes place by fragmentation or by formation of different types of spores. Zoospore formation is rare. Sexual reproduction can be isogamous, anisogamous, or oogamous. Asexual reproduction can occur by flagellated zoospores produced in zoosporangia.

15. Correct option: (2) a-iv, b-ii, c-i, d-iii

Zoospores are motile asexual spores having two polar flagella. Fucus produce oogamous spores with single large female spore and small male spores. Chlamydomonas produces isogamous or similar types of motile gametes. Spirogyra produces anisogamous spores.

16. Correct option: (C) Both (A) and (B)

Certain marine brown and red algae produce large amounts of hydrocolloids (water holding substances), e.g., algin (brown algae) and carrageen (red algae) which are used commercially.

17. Correct option: (C) Gelidium and Gracilaria

Gracilaria has become the preferred seaweed for making food grade agar because it has been successfully cultivated in Chile and Indonesia. Gelidium continues to be the preferred seaweed for making bacteriological and pharmaceutical grade agars and agarose.

18. Correct option: (A) Chlorophyceae

The members of Chlorophyceae are commonly called green algae. The plant body may be unicellular, colonial, or filamentous. They are usually grass-green due to the dominance of pigments chlorophyll a and b.

19. Correct option: (D) Cellulose, pectose

Cell wall of green algae is thin, transparent and firm and consists of outer pectic and inner cellulosic layers. It is smooth but in most of species it gets thickened at the anterior end to form an apical papilla.

20. Correct option: (A) Both assertion (A) and Reasoning (R) are correct and (R) explains (A)

Pyrenoids are characteristically found in the chloroplast of algae. They are one or more storage bodies present mainly in Chlorophyceae (green algae). Pyrenoids contain protein and starch. RuBisCO enzyme is also present in pyrenoids. It helps in CO_2 fixation in plants.

21. Correct option: (A) Both assertion (A) and Reasoning (R) are correct and (R) explains (A)

Carotenoids, xanthophylls, chlorophyll a, and c are all present in Phaeophycean. Depending on the concentration of xanthophyll pigments, members of the phaeophycean family exhibit colour variations ranging from olive green to various hues of brown.

22. Correct option: (D) Both 1 and 2

Phaeophyceae are commonly known as brown algae. They possess chlorophyll a, c, xanthophylls, and carotenoids. Food is stored in the form of complex carbohydrates, mostly in the laminarin or mannitol form.

23. Correct option: (C) b and d

The members of Phaeophyceae or brown algae are found primarily in marine habitats. They show great variation in size and form. In the class phaeophyceae of algae, the plant body is usually attached to the substratum by a holdfast, and has a stalk, which is termed the stipe, and a leaf-like photosynthetic organ called the frond.

24. Correct option: (B) Statement- I is correct and II is also correct

In Algae, there are 3 types of sexual reproduction, Isogamy, Anisogamy, and Oogamy. In Isogamy male and female gametes have similar morphology. They have similar shape and size. In Anisogamy male and female gamete have different morphology. In Oogamy female gamete is significantly larger than the male gamete. In this, the female gamete is non motile and male gamete is motile. Gamete union can occur in the oogonium or water (oogamous species).

25. Correct option: (D) Laminaria, Ectocarpus, Sargassum

Brown algae are a group of algae belonging to class Phaeophyceae. They are named due to their colour, which varies from brown to olive green. They contain seaweeds, some of the notable examples are Ectocarpus, Fucus, giant kelps, Sargassum, etc.

26. Correct option: (C) Laminaria and sargassum

Laminaria and Sargassum are the brown algae (Phaeophyceae) and are cultivated as a crop.

27. Correct option: (B) red algae

Due to the dominance of red pigment r-phycoerythrin in the algae, the body appears to be red. These red algae are known as Rhodophyceae.

28. Correct option: (A) Both Assertion (A) and (R) are correct and (R) explains(A)

The red algae flourish and occur in abundance at great depths of sea where other plants do not occur in abundance. The red thalli of most of the red algae are multicellular. Some of them have complex body organisation.

29. Correct option: (C) Holdfast

In the class phaeophyceae of algae, the plant body is usually attached to the substratum by a holdfast, and has a stalk, which is termed the stipe, and a leaf-like photosynthetic organ called the frond.

30. Correct option: (A) Zoospores

Asexual reproduction in most brown algae is by biflagellate zoospores that are pear-shaped and have two unequals laterally attached flagella.

LEVEL-II

1. Correct option: (B) Taxonomy

The science of identification, naming, and classification of organisms is known as taxonomy.

2. Correct option: (C) Carolus Linnaeus

Linnaeus' gift to science was taxonomy: a classification system for the natural world to standardize the naming of species and order them according to their characteristics and relationships with one another.

3. Correct option: (B) Aristotle

Aristotle is known as the father of zoology. Zoology is a branch of biology that deals with animals' life, evolution, anatomy, physiology, and behaviour. Aristotle is also known as the father of biology; he was an ancient Greek philosopher.

4. Correct option: (C) Whittaker

Five kingdom of classification was proposed by RH Whittaker in 1969. He divided all living organisms into five kingdoms based on following features: Complexity of cell structures. Complexity of body structures.

5. Correct option: (D) species

In this system of classification, kingdom is always ranked the highest followed by division, class, order, family, genus, and species.

6. Correct option: (A) Cat

The Felidae family, comprised of 42 species including the domestic cat, is widely divergent in geographic range and phenotype, with body sizes ranging from 1.5 to 300 kg.

7. Correct option: (A) Thallus

Plants whose body is not differentiated into distinct roots, stems and leaves are called thallophytes.

8. Correct option: (A) Phycology

Phycology, also called algology, the study of algae, a large heterogeneous group of chiefly aquatic plants ranging in size from microscopic forms to species as large as shrubs or trees.

9. Correct option: (D) Lichens

Lichens: Symbiotic relationship between algae and fungi is called lichens. The fungi provide water and minerals to the alga, while the algae perform photosynthesis and supplies food in the form of sugars to the fungi.

10. Correct option: (B) Bryophytes

Bryophytes are called the amphibians of the plant kingdom. Amphibians are those organisms which live on both land and in water. Bryophytes are called amphibians of the plant kingdom because these plants need both water and soil for their survival.

11. Correct option: (A) Sporophyte

The main plant body of the bryophyte is haploid. It produces gametes, hence is called a gametophyte. The male and female gametes fuse together to form zygote. The zygote produces a multicellular body called a sporophyte which is diploid.

12. Correct option: (C) E. Haeckel

The term “Pteridophyta” was given by Ernst Haeckel in the year 1866. Pteridophytes are considered as the first land plants with a well-developed vascular system. The Greek word “Pteron” means feather and “Phyton” means plants.

13. Correct option: (A) Heterosporous

Pteridophytes like Selaginella and Salvinia produce two kinds of spore (microspores and macrospores). They are known as heterosporous.

14. Correct option: (A) Gymnosperms

Gymnosperms are non-flowering plants and bear naked seeds. For example - Pine.

15. Correct option: (C) Leaves

The leaves in gymnosperms are well adapted to withstand extremes of temperature, humidity, and wind. In conifers, the needle-like leaves reduce the surface area. Their thick cuticle and sunken stomata also help to reduce water loss.

16. Correct option: (A) Gymnosperms

Gymnosperms are the seed-producing plants, but unlike angiosperms, they produce seeds without fruits. These plants develop on the surface of scales or leaves, or at the end of stalks forming a cone-like structure. Pinus and Cycas are examples of gymnosperms.

17. Correct option: (D) Angiosperms
Angiosperms are plants that produce flowers and bear their seeds in fruits. They are the largest and most diverse group within the kingdom Plantae, with about 300,000 species.
18. Correct option: (B) 2
There are two types of root systems: taproots and fibrous roots (also known as adventitious roots) in angiosperms present in monocots and dicots respectively.
19. Correct option: (B) 2
The dicotyledons, also known as dicots (or, more rarely, dicotyls), are one of the two groups into which all the flowering plants (angiosperms) were formerly divided. The name refers to one of the typical characteristics of the group: namely, that the seed has two embryonic leaves or cotyledons.
20. Correct option: (D) flower
Angiosperms are commonly known as flowering plants. Flowers are the reproductive parts of the plant as it produces the specialised reproductive cells called gametes.
21. Correct option: (B) Pistil
The female part is the pistil. The pistil usually is in the centre of the flower and is made up of three parts: the stigma, style, and ovary. The stigma is the sticky knob at the top of the pistil. It is attached to the long, tubelike structure called the style.
22. Correct option: (B) Pistil
Pistil: The part of the flower which contains the ovary is called Pistil. It comprises style, stigma, and ovary. The ovary in flowering plants is a part of the female reproductive structure of the gynoecium.
23. Correct option: (A) Pollination
Pollination is the act of transferring pollen grains from the male anther of a flower to the female stigma.
24. Correct option: (D) Zygote
One male gamete fuse with the egg cell to form the zygote that gives rise to the embryo.
25. Correct option: (B) pollen grains
Anther, in flowering plants, the part of a stamen that produces and contains pollen. Each anther is generally borne at the tip of a long slender stalk known as a filament and consists of two lobes that each house a pair of pollen sacs (microsporangia) that produce pollen for pollination.

Animals

LEVEL-I

1. Correct option: (B) & (C) Mollusca and Annelida
The body cavity between the body wall and the gut wall, lined by mesoderm, is called coelom. Animals possessing coelom are called coelomates, e.g., annelids, molluscs, arthropods, echinoderms, hemichordates, and chordates.
2. Correct option: (D) Aschelminths
Bilateral Symmetry refers to the division of a body into two equal halves in such a way that both halves are mirror images of each other or the same on both sides of an axis. Triploblastic refers to the organisms whose body is derived from three germ layer. Ctenophores are biracially symmetrical and diploblastic, whereas some species are triploblastic. Sponges are asymmetrical and diploblastic. Coelenterates are radially symmetrical and diploblastic. Aschelminths are bilaterally symmetrical and triploblastic.

3. Correct option: (A) Sponges

Phylum Porifera members also known as sponged have a water transport or canal system. Water enters through the minute pores of sponges called ostia in the body wall into a central cavity called spongocoel, from where it goes out through the osculum.

4. Correct option: (B) Echinodermata

Echinoderms have a water-driven tube system that the members of this phylum use for their locomotion. This system is also known as the water vascular system. It also helps in food capturing.

5. Correct option: (D) Eight

The body of a ctenophore typically bears eight external rows of ciliated comb plates, which help in locomotion.

6. Correct option: (C) Excretion and Osmoregulation

Flame cells are present in the phylum Platyhelminthes (flatworms). Flame cells are the specialized excretory cells that help in excretion in animals. These cells work like kidneys and help in removing waste materials from the body of the animals.

7. Correct option: (C) Echinodermata

8. Correct option: (A) A- Radially symmetrical B- Bilaterally symmetrical

Echinoderms are a phylum of marine animals. The adults are recognizable by their (usually five-point) radial symmetry, whereas the larvae are bilaterally symmetrical.

9. Correct option: (A) Hemichordate

Excretory organ in animals belonging to the phylum Hemichordate is the proboscis gland. A proboscis is an elongated appendage from the head of an animal, either a vertebrate or an invertebrate.

10. Correct option: (D) Ventral cord, nerve cord

Chordates are the group of animals which possess a notochord. They have a tubular hollow and dorsal nerve cord. The notochord is the rod like structure. It is made up of cartilage and extends along the length of the body. In certain groups, the notochord forms the tail like structure which helps in swimming in case of fishes. They have a ventral heart. There is presence of pharyngeal slits. There is an endostyle and a post anal tail for at least some period of their life.

11. Correct option: (A) Torpedo

Torpedoes are also known as electric ray. They have electric organs present that act as a defines mechanism against predators.

12. Correct option: (D) Limulus

Limulus commonly known as king crab belongs to phylum Arthropoda. It is a living fossil and large sized marine arthropod.

13. Correct option: (D) Choanocytes

Choanocytes (also known as "collar cells or flagellated cells ") are cells that line the spongocoel. The water current of the body of sponge is maintained by the movement of flagella of choanocytes.

14. Correct option: (A) i & ii

Parapodia are present in some annelid worms. These are a pair of flattened flesh lobes bearing numerous bristles. Radula (rasping organ) in molluscs is a structure involved in feeding.

15. Correct option: (A) Roundworms are pseudocoelomates

Platyhelminthes or flatworms are acoelomates i.e., they lack a coelom. Molluscs and arthropods are coelomates as they have a true coelom derived from the germ layer mesoderm. Aschelminths or roundworms are pseudocoelomates as the coelom is not derived from the mesoderm.

16. Correct option: (A) Both (A) and (R) are true and '(R)' is correct explanation of 'A'
Tetrapod is a superclass of animals that includes all limbed vertebrates that primarily live on land. Fishes are adapted to live in aquatic habitats. As a result, instead of limbs, they have paired fins that assist them in swimming, balance, stability, and steering. So, fishes are not considered as tetrapod's due to the presence of fins.

17. Correct option: (A) Catfish
Catfish is true fish having prominent barbels without scales. Jellyfish is a free-swimming marine animal that has an umbrella-shaped body. Cuttlefish is a mollusc. Silverfish is a wingless insect.

18. Correct option: (A) Both (A) and (R) are true and '(R)' is correct explanation of 'A'
Amphibian eggs do not have a hard protective shell-like bird or reptile eggs. Frogs and toads lay eggs in jelly-like masses, and unless the eggs are covered with water, they will dry up and die. This is why amphibians breed in water. The reproduction occurs by the process of external fertilization. The females release their eggs into the water which are then fertilized by the male sperm.

19. Correct option: (C) Crocodile
Crocodile is the only exceptional animal that belongs from class Reptiles and has a four-chambered heart.

20. Correct option: (A) Ectotherms
Ectotherms are cold-blooded animals that depend on the temperature of surroundings to change their own body temperature to survive. Endotherms are warm-blooded animals which able to maintain constant body temperature, even in extreme environments. Psychrotherms are animals that lives in the cold while thermophiles are animals that live in hot springs.

21. Correct option: (A) Both (A) and (R) are true and '(R)' is correct explanation of 'A'
Mammals and Aves are homeothermic animals, meaning they can maintain a constant body temperature. They have evolved various mechanisms to do so, sweating and shivering are some of the most common ones. Hence, they are able to cope up in extreme climatic conditions.

22. Correct option: (C) Bats
Echolocation is a technique used by bats, dolphins and other animals to determine the location of objects using reflected sound.

23. Correct option: (D) Viviparity is not found in species of lizards and snakes
Majority of mammals are viviparous, which means they give birth to young ones. The egg is retained and implanted in the mother's uterus, the eggshell does not form, and a placenta forms that serves to supply nourishment and gas exchange between the mother and embryo. Most reptiles and snakes give birth to live young. They are viviparous animals. They produce living young instead of eggs from within the body.

24. Correct option: (D) All of the above
The common ancestor of birds and mammals must have occurred at the split of the Sauropsida (Reptiles) and Synapsida (ancestors of mammal-like reptiles). The amnion, yolk sac, chorion, and allantois are four extraembryonic membranes found in reptile, avian, and mammalian embryos.
In an amniotic sac filled with amniotic fluid, the embryo is protected by amnion. Outer skin covering of reptiles, mammals and birds are watertight.

25. Correct option: (C) Platypus
Platypus (Ornithorhynchus) and Echidna (Tachyglossidae) are oviparous mammals. In oviparous animals, internal fertilization occurs but development takes place outside the body.

26. Correct option: (A) Hollow body

A sponge's body is hollow and is held in shape by the mesohyl, a jelly-like substance made mainly of collagen and reinforced by a dense network of fibres also made of collagen. The inner surface is covered with choanocytes, cells with cylindrical or conical collars surrounding one flagellum per choanocyte.

27. Correct option: (B) Coelenterate

Coelenterate animals are diploblastic animals. The mesoglea layer lies between the ectoderm and the endoderm. Mesogloea is helpful in providing buoyancy. Mesoglea is a non-cellular layer.

28. Correct option: (C) Physalia

The Portuguese-man-of-war, or the Physalis physalis, also known as bluebottle, is a marine cnidarian of the family Physaliidae. Its venomous tentacles can deliver a painful sting.

29. Correct option: (D) Mouth, pharynx, intestines, and Anus

The digestive system of a nematode is made up of three main parts: the stomodeum, the intestine, and the proctodaeum. The stomodeum is the beginning of the digestive tract where we find the mouth opening, esophagus (also called the pharynx), and the buccal cavity (the area inside the mouth).

30. Correct option: (D) Insects

Hemocoel is the cavity between the organs of arthropods and molluscs, through which the blood circulates.

31. Correct option: (B) Tail region

Urochordates, like Herdmania have notochord and nerve cord present only in nerve stages. Body is covered by a test made up of tunicin. Notochord is present only in tail. Tadpole undergoes retrogressive metamorphosis and notochord, and nerve cord is absent in adults, which is lost during metamorphosis.

32. Correct option: (D) Tortoise

A tortoise is not a true amphibian animal; it is a reptile. A tortoise can live on land as well as in water but its body is covered by the shell (provides protection), dorsally by carapace and ventrally by plastron. The presence of protective shells is not a characteristic feature of true amphibians. Amphibians are vertebrates that live on both land and water. They respire through gills and moist body surface but on land, they respire through lungs. Frogs, toads and salamanders are examples of amphibians.

33. Correct option: (C) Ornithorhynchus

Platypus (Ornithorhynchus) and Echidna (Tachyglossidae) are oviparous mammals. In oviparous animals, internal fertilization occurs but development takes place outside the body.

34. Correct option: (A) 4, 2, 3, 1, 4

Water canal system is a characteristic feature of phylum Porifera. The body of the poriferans have pores called ostia. The body has only one opening called the osculum. There is a continuous water movement from the exterior to the interior of the body of the poriferans. Water flows through the ostia into the spongocoel and goes out through the osculum. This helps in food gathering, exchange of gases and removal of waste.

35. Correct option: (D) 3 and 4

Most flatworms are hermaphrodite that is they posses both male and female sex organs. Fertilization is internal in case of them and occurs through cross-fertilization. Most of the flatworms have a direct mode of development. And polyembryony is quite common amongst them.

36. Correct option: (C) A is true, R is false

Ascaris belong to the phylum aschelminths in which fertilization is internal and development may be direct or indirect. Ascaris usually lay eggs and thus are oviparous.

37. Correct option: (D) 1, 2, and 3

Nematodes are dioecious with separate male and female individuals. They possess an internal fertilization. Nematodes may produce eggs (oviparous) or larvae (viviparous). Some lay eggs

containing larvae which immediately hatch out (ovoviviparous). The life cycle consists typically of four larval stages and the adult form. The most common life cycle pattern among parasitic nematodes is the direct one.

38. Correct option: (D) They have a circulatory system

Flatworms breathe through their moist body surface (skin) and diffusion of oxygen takes place directly. By this diffusion process, each cell can meet its oxygen requirements thus there is no need for the circulatory system to circulate oxygen and nutrients.

39. Correct option: (C) They are protostomes

The echinoderms and the chordates are deuterostomes, while all other invertebrates are protostomes. The main difference between protostomes and deuterostomes is that the blastopore in protostomes is developed into a mouth while the blastopore in deuterostomes is developed into an anal opening.

40. Correct option: (B) Hairy skin and mammary glands

Two characteristics are used to define the mammal class. They are mammary glands and body hair (or fur). Female mammals have mammary glands.

LEVEL-II

1. Correct option: (B) Two

Animal kingdom can be broadly classified into two major groups i.e., vertebrates and invertebrates.

2. Correct option: (C) Chordata

Animals with notochord are called chordates. In some chordates, the notochord is modified as the vertebral column. Such animals are called vertebrates.

3. Correct option: (A) 2

Diploblastic animals have two cell layers and they are outer ectoderm and inner endoderm. These animals don't have the mesoderm whereas they have middle mesoglea which is an undifferentiated layer.

4. Correct option: (D) Sponges

The phylum Porifera comprises the sponges. Sponges are simple invertebrate animals that live in aquatic habitats.

5. Correct option: (D) Ctenophores

Ctenophora is a phylum including the animals that live in marine waters worldwide. Their most distinctive feature is the combs groups of cilia, which they use for swimming. They are the largest animals that swim by the means of cilia. Comb jellies belong to phylum Ctenophora.

6. Correct option: (A) External

Ctenophores reproduce sexually. Fertilisation occurs outside the body. Development is indirect.

7. Correct option: (B) Excretion

Excretion and osmoregulation in Platyhelminthes occur with the help of a protonephridia system having flame cells.

8. Correct option: (A, B) Aschelminths / Nematode

Incomplete digestive tract i.e., anus is absent, found in Platyhelminthes and Ctenophora Aschelminths/ Nematode and Annelida have both mouth and anus, which have complete digestive tract.

9. Correct option: (B) Nephridia

Nephridia is an excretory organ that eliminates nitrogenous waste from the body cavity and removes them through the pores outside the body. Numerous species have a couple of nephridia in each portion.

10. Correct option: (C) Arthropoda

The organ system level of organization is when two or more organs work together for a specific function. Annelids, Arthropods, Molluscs, Echinoderms and Chordates have this level of organization.

11. Correct option: (C) Arthropoda

Insects belong to the phylum Arthropoda. Arthropods move with the help of jointed appendages. The word "Arthropod" means "jointed feet".

12. Correct option: (A) Mollusca

Mollusca is the second largest phylum in the animal kingdom. Molluscs have the following characteristics: They are bilaterally symmetrical. They are triploblastic; have presence of three germ layers. This phylum includes a wide variety of marine, freshwater, and terrestrial species. It includes about 100,000 living species.

13. Correct option: (C) Sensory

The anterior head region has sensory tentacles in molluscs.

14. Correct option: (A) Excretory

The excretory organs of Mollusca are metanephridia. Metanephridia filters waste from internal body fluid.

15. Correct option: (B) Mollusca

The simplest sort of respiratory system found in molluscs is cutaneous respiration. The dorsal surface of the body is covered in papillae in the majority of the Aeolididae species. The papillae differ in size and are connected to the heart by veins.

16. Correct option: (B) Gill slits

Hemichordates consists of a small group of worm-like marine animals with an organ-system level of organization. They are no chordates. The circulatory system is of open type. The blood flows to a cavity known as hemocoel which bathes the organs directly with oxygen and nutrients and there is no distinction between blood and interstitial fluid. They respire with help of gill slits.

17. Correct option: (B) Excretory

Proboscis gland is the excretory organ of hemichordate.

18. Correct option: (A) hemichordate

Balanoglossus belongs to the phylum Hemichordate, the animals in this group are also called acorn worms.

19. Correct option: (D) Both are true

Animals in the phylum Chordata share four key features that appear at some stage during their development: a notochord, a dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail. In some groups, some of these are present only during embryonic development.

20. Correct option: (C) 7

Broadly Vertebrata is divided into Agnatha and Gnathostomata; Class Cyclostomatous is under Agnatha and other 6 are under Gnathostomata including Chondrichthyes, Osteichthyes, Amphibia, Reptilia, Aves, and Mammalia.

21. Correct option: (D) Kidney

Cyclostomes have a pair of mesonephric kidneys that make up the excretory system of the organism.

22. Correct option: (A) Amphibians

The first amphibians evolved from a lobe-finned fish ancestor about 365 million years ago. They were the first vertebrates to live on land, but they had to return to water to reproduce. This meant they had to live near bodies of water.

23. Correct option: (C) 3

Heart of Amphibians: It has three chambers – one ventricle and two atria.

24. Correct option: (D) 4

Reptiles generally have a three-chambered heart, while crocodile is an exception with a four-chambered heart. There are two atrium and two ventricles in the heart forming the four chambers.

25. Correct option: (C) Lungs

The avian respiratory system exchanges oxygen and carbon dioxide between the gas and the blood utilizing a relatively small, rigid, flow-through lung, and a system of air sacs that act as bellows to move the gas through the lung.

26. Correct option: (D) 4

Mammal hearts have four chambers (two atria and two ventricles).

27. Correct option: (C) Both A and B

Generally, all mammals are viviparous that is they give birth to their young ones, except some monotremes like Platypus and Echidna that can lay eggs and are oviparous.

28. Correct option: (A, C) Aves, Mammalia

Aves are warm-blooded animals and have a four-chambered heart. All mammals and birds can generate this internal heat and are classed as homoiotherms, or warm-blooded animals. Normal temperatures for mammals range from 97° F to 104° F. Most birds have a normal temperature between 106° F and 109° F.

29. Correct option: (B) Cold blooded

Reptiles are cold-blooded, or ectothermic, animals. This means that they cannot produce heat in their own bodies and have to rely on their surroundings to keep warm. This is why most reptiles are found in hot or warm climates.

2. The Fundamental Unit of Life

Solutions

LEVEL – I

1. Correct option: (D) All of these

The cytoplasm is kept apart from the outside environment by the cell membrane, sometimes referred to as the plasma membrane. It is composed of a phospholipid bilayer with proteins incorporated into it. The hydrophobic tails of the lipids are facing each other and the hydrophilic heads are facing outward in the phospholipid bilayer, which is a double layer of lipids. The cell membrane may be selectively permeable because to its structure.

Many essential functions are carried out by the cell membrane, including the following:

- controlling the flow of materials into and out of the cell.
- putting up a wall between the cell and its surroundings.
- defending the cell against hazardous elements.
- maintaining the interior environment of the cell.

2. Correct option: (C) Robert Brown

Scottish botanist Robert Brown is well recognized for identifying the nucleus in plant cells in 1831. He was looking at plant cells under a microscope when he saw a tiny, solid mass in the middle of the cell. The "smallest resolvable part" of the cell, this structure was given the term "nucleus" by the scientist.

Nucleus, which meaning "kernel" or "seed" in Latin, is where the term "nucleus" derives from. Given that the nucleus is the core of the cell and houses the genetic material for the cell, the name "nucleus" is appropriate.

Camillo Golgi was an Italian physician and biologist who discovered the Golgi apparatus in 1898.

Matthias Schleiden was a German botanist who, along with Theodor Schwann, is credited with the discovery of the cell theory.

Rudolf Virchow was a German physician and pathologist who coined the term "cell theory".

3. Correct option: (D) Mitochondria

The mitochondria are the engine of the cell. Eukaryotic cells—those having a well-defined nucleus and numerous other membrane-bound structures—have mitochondria, which are membrane-bound organelles. These organelles are sometimes referred to as the "powerhouses" of the cell because they are crucial to the process of cellular respiration, which produces adenosine triphosphate (ATP), the main source of energy for the cell.

Plastids are organelles that are found in plant cells. They are responsible for photosynthesis, which is the process by which plants convert light energy into chemical energy.

Lysosomes are organelles that are found in all cells. They are responsible for breaking down cellular waste products.

RER stands for rough endoplasmic reticulum. It is a type of endoplasmic reticulum that is covered in ribosomes. Ribosomes are responsible for making proteins.

4. Correct option: (B) Cell wall

A plant cell's cell wall is the additional layer that exists outside of the cell membrane. The cell wall, which is mostly made of cellulose, gives the cell structural support and defense, keeping it from bursting or collapsing under diverse environmental circumstances. Animal cells, which lack this stiff outer layer and are therefore more flexible in shape and movement, are distinguished from plant cells by the absence of the cell wall, which is a distinguishing trait of plant cells.

Plasma membrane is the outermost layer of all cells, including plant cells. It is a thin, flexible layer that protects the cell and controls what enters and leaves the cell.

Epidermis is the outer layer of plant tissue. It is made up of cells that are tightly packed together. The epidermis protects the plant from the environment and helps to regulate water loss.

Biological membrane is a general term for the outer layer of all cells. It is made up of lipids and proteins.

5. Correct option: (B) Prokaryotes

Blue-green algae, often known as cyanobacteria, are prokaryotic organisms that fall within the realm of bacteria. Prokaryotes are unicellular creatures that are organelle-free and devoid of a genuine nucleus. Instead, a single circular DNA molecule found in the nucleoid region houses all of their genetic information. Similar to plants, cyanobacteria are photosynthetic microorganisms that can perform photosynthesis to create energy and oxygen. They contribute to the ecosystem's ability to produce oxygen and fix nitrogen, which makes them crucial for many aquatic habitats.

Eukaryotes, on the other hand, do contain a nucleus and other organelles that are attached to membranes.

Biological entities classified as mesokaryotes share traits with both prokaryotes and eukaryotes. Although their nucleus is there, a double membrane does not encircle it.

Two nuclei can be found in each cell of a species of organisms called dikaryotes.

6. Correct option: (C) Nucleoplasm

Eukaryotic cells have a unique fluid called nucleoplasm within their nucleus. It encloses and encircles the nucleus and chromosomes, among other genetic components. This liquid is crucial for preserving the nucleus's integrity and structural integrity. Proteins, enzymes, and other substances essential for nuclear and gene control are abundant in the nucleoplasm. In addition to being essential for the efficient operation and regulation of genetic information within the nucleus, it plays a significant role in coordinating the actions of the cell.

Although they are not exactly correct, the phrases "nuclear fluid" and "karyoplasm" are occasionally used interchangeably with nucleoplasm. While karyoplasm refers to the entire nucleus's contents, including the chromosomes, nucleoli, and nucleoplasm, nuclear fluid is a broader phrase that may be used to describe any fluid found in the nucleus.

So, the answer to the question is C) Nucleoplasm.

7. Correct option: (D) Both a & b

Lysosomes are known as both "A) Digestive vesicles" and "B) Suicidal bags."

The membrane-bound organelles known as lysosomes have enzymes that may degrade cellular trash, foreign objects, and even fragments of the host cell. They are sometimes referred to as "suicide bags" because the enzymes they contain have the ability to kill cells if they burst.

Lysosomes are also referred to as "digestive vesicles" since they are crucial for the breakdown of cellular waste. Lysosomes contain hydrolytic enzymes that help break down and digest various cellular materials, such as waste products, damaged organelles, and foreign particles.

8. Correct option: (A) 150 times

In comparison to the nucleus, mitochondria are around 150 times smaller. The biggest organelle in a cell is the nucleus, which houses the DNA for that cell. Because they are significantly smaller than the nucleus, mitochondria provide the majority of the energy in a cell. They play a crucial role in cellular respiration, generating energy in the form of ATP through a series of biochemical reactions.

The size of a mitochondrion can vary depending on the type of cell, but they are typically about 1-2 micrometers in length and 0.5-1 micrometers in diameter. The nucleus, on the other hand, is typically about 5-10 micrometers in diameter.

9. Correct option: (B) Endoplasmic reticulum

A network of membranes called the endoplasmic reticulum (ER) is present in the cytoplasm of eukaryotic cells. It is in charge of a number of processes, such as detoxification, lipid synthesis, and protein synthesis.

Rough endoplasmic reticulum (RER), which has ribosomes attached to its surface, and smooth endoplasmic reticulum (SER), which is ribosome-free, are two different forms of ER. The endoplasmic reticulum's network of membranes offers a significant surface area for several cellular activities and makes communication between diverse parts of the cell easier.

10. Correct option: (D) Both a & c

Plant and algal cells include double-membrane organelles called plastids. They may participate in photosynthesis and are in charge of the synthesis and storage of food. Chloroplasts, chromoplasts, and leucoplasts are the three primary categories of plastids. The most typical kind of plastids are chloroplasts. They contain chlorophyll, the substance that gives plants their green color. Photosynthesis, the process by which plants turn light energy into chemical energy, is carried out by chloroplasts.

Chromoplasts are plastids that include pigments besides chlorophyll. Fruits and flowers have their distinctive colors thanks to these pigments. Pigments are created and stored by chromoplasts.

11. Correct option: (C) Both lipids and proteins

The cell membrane, also known as the plasma membrane, is composed of both lipids and proteins. It is a thin, flexible barrier that surrounds the cell and separates its internal environment from the external environment.

The main components of the cell membrane are phospholipids, which form a lipid bilayer, with hydrophilic heads facing outward and hydrophobic tails facing inward. Proteins are embedded within this lipid bilayer, either spanning the entire membrane (integral proteins) or loosely associated with the surface (peripheral proteins).

Together, the lipids and proteins in the cell membrane play essential roles in maintaining the cell's structure, regulating the movement of molecules in and out of the cell, and facilitating cell communication and recognition.

Cellulose, mentioned in option D, is a carbohydrate found in the cell walls of plants but not a part of the cell membrane.

12. Correct option: (A) Nucleus

The component of the cell that contains the genetic code is called the nucleus. The DNA of the cell, which holds the genetic instructions for the cell's construction, function, and reproduction, is what gives it the nickname "control center" of the cell. The nucleolus, which is located inside the nucleus and is important in ribosome production, is there.

The nucleolus is a small, dense region within the nucleus that is involved in the production of ribosomes. Ribosomes are the organelles that synthesize proteins.

Ribosomes are not found in the nucleus, but they are produced there. They are then exported to the cytoplasm, where they carry out protein synthesis.

The Golgi complex is a stack of flattened sacs that is involved in the processing and packaging of proteins. It is not involved in the storage of genetic information.

13. Correct option: (D) Both Bacterium and Cyanobacteria

Cyanobacteria and bacteria are both prokaryotic organisms. Cells without a nucleus or other membrane-bound organelles are known as prokaryotic cells. Their genetic material occurs in the cytoplasm as a single circular DNA molecule rather than being contained

within a membrane-bound nucleus. Prokaryotic creatures include bacteria and cyanobacteria

A eukaryotic cell is Paramecium sp. A nucleus and other membrane-bound organelles are features of eukaryotic cells. They are present in all three domains of life, with the exception of Archaea, and are more complicated than prokaryotic cells.

Hence the answer is D) Both Bacterium and Cyanobacteria

14. Correct option: (A) Endoplasmic reticulum

Intracellular transport does definitely entail the endoplasmic reticulum (ER). The enormous network of membranes in the cell is responsible for transporting proteins and other materials. The ribosome-filled rough ER makes proteins and aids in their folding and transport. In addition to helping with the transit of these chemicals, the smooth ER is engaged in lipid production. As a result, the endoplasmic reticulum is crucial for the movement of proteins and lipids inside the cell.

A network of membranes called the endoplasmic reticulum (ER) is present in the cytopl asm of eukaryotic cells.

It is in charge of a number of processes, such as detoxification, lipid synthesis, and protein synthesis. In the process of intracellular transport, the ER is crucial.

The Golgi complex receives and processes the proteins that are produced on the riboso mes of the rough ER.

The proteins are subsequently put into vesicles by the Golgi complex and sent to variou s cell locations.

So, the answer to the question is A) Endoplasmic reticulum. The endoplasmic reticulum participates in intracellular transport.

15. Correct option: (B) Lipids

Lipid synthesis is carried out by the smooth endoplasmic reticulum (SER). Under a microscope, it appears smooth because its surface is devoid of ribosomes. Lipid metabolism, which includes the production of numerous lipids including phospholipids, cholesterol, and steroid hormones, is one of its primary tasks.

The smooth ER also stores calcium ions, which are necessary for cellular signaling and muscle contraction, and is in charge of detoxifying certain medications and poisons from the cell. The smooth endoplasmic reticulum (ER) is an expert in lipid-related activities, whereas the rough endoplasmic reticulum (RER) is predominantly involved in protein synthesis.

16. Correct option: (A) Smooth endoplasmic reticulum

The smooth endoplasmic reticulum (SER), found in vertebrate liver cells, is essential for the detoxification of several toxins and medicines. It has cytochrome P450 enzymes, which detoxify medications, poisons, and other hazardous substances. These enzymes change these chemicals, making them more water-soluble and simpler for the body to remove via urine or bile. The smooth ER found in liver cells is an important part of the body's detoxification system, which depends on the liver for metabolic and detoxifying functions.

An endoplasmic reticulum that contains ribosomes connected to it is known as the rough endoplasmic reticulum (RER). It contributes to the creation of proteins.

Organelles with a membrane around them, lysosomes have enzymes that digest cellular trash.

Organelles with a membrane around them, vacuoles store waste, nutrients, and water.

17. Correct option: (B) 150 – 3000

Depending on the kind of cell and its level of activity, different types of cells have different numbers of mitochondria. For instance, because muscle cells require a lot of energy to

contract, they have several mitochondria. Red blood cells, on the other hand, lack mitochondria since they do not require energy to function.

There are around 150–3000 mitochondria per cell on average. Some cells, nevertheless, contain more or fewer mitochondria than this. For instance, although a human sperm cell only has one mitochondrion, a human liver cell has roughly 2000 mitochondria.

The cellular energy currency, ATP, is created by mitochondria. Therefore, a cell's energy needs are correlated with the quantity of mitochondria present. Muscle cells and red blood cells contain more mitochondria than cells with lower energy requirements like white blood cells.

18. Correct option: (B) Photosynthesis

Chloroplasts perform photosynthesis as their main activity. The function of chloroplasts, which are organelles present in plant cells and certain algae, is to absorb light energy and transform it into chemical energy in the form of glucose. In this procedure, known as photosynthesis, light energy is absorbed by chlorophyll, a green pigment found in chloroplasts. Utilizing the energy of the collected light, glucose and oxygen are created by combining carbon dioxide and water.

Respiration is the process by which cells use oxygen to break down carbohydrates and other molecules to produce energy. Nutrition is the process by which cells take in nutrients from the environment and use them to build and maintain the cell. Transport is the process by which cells move materials around inside the cell and between cells.

So, the answer is B) Photosynthesis. The primary function of chloroplasts is to carry out photosynthesis.

19. Correct option: (C) Both Schleiden & Schwann

Both Theodor Schwann and Matthias Schleiden put out the cell idea. They were eminent scientists who contributed significantly to our understanding of cells and living things in the 19th century.

German botanist Matthias Schleiden made the claim that plants are made up of cells in 1838. Schleiden's theories were expanded upon by another German scientist, Theodor Schwann, who in 1839 postulated that animal tissues also contain cells

Together, their efforts laid the groundwork for the cell hypothesis, which contends that since all living things are made up of cells, they are life's fundamental building blocks.

The third scientist, Rudolf Virchow, also made substantial contributions to cell theory, but he is best remembered for the phrase "Omnis cellula e cellula" (every cell derives from another cell), which emphasizes the idea of cell division and the continuity of life.

20. Correct option: (A) Nucleoplasm

Nucleoplasm refers to the liquid that fills a eukaryotic cell's nucleus. The genetic material of the cell, including the nucleolus and chromosomes, is encased in and contained by nucleoplasm, a gel-like substance. The nuclear membrane, which encloses it and divides it from the cytoplasm, surrounds it.

The fluid that surrounds the cell's nucleus but is still within, known as the cytoplasm (option B), includes a variety of organelles and is the scene of several cellular activity.

Historically, the live components of a cell, which include the cytoplasm and nucleus, were referred to as protoplasm (option C).

The term "germplasm" (option D) refers to the genetic material in the germ cells, which are in charge of transmitting genetic information to progeny.

21. Correct Matching is: a-T;b-P;c-S;d-Q;e-R

a) Robert Hook – T) Living cell

The word "cell" was first used by British scientist Robert Hooke in 1665. In cork tissue, he noticed cells, which he characterized as "little boxes".

b) Robert Brown – P) Nucleus
The nucleus was found in 1831 by Scottish botanist Robert Brown. He examined plant cell nuclei and referred to them as "highly refractive bodies."
c) Schwann – Q) Zoologist
Theodor Schwann was a German zoologist who proposed that all animal cells are made of cells. He published his findings in 1839.
d) Schleiden – Q) Botanist
Matthias Schleiden was a German botanist who proposed that all plant cells are made of cells. He published his findings in 1838.
e) Camillo Golgi – R) Golgi apparatus
Camillo Golgi was an Italian physician and histologist who discovered the Golgi apparatus in 1898. He described the Golgi apparatus as a "network of canals".

22. Correct Matching is: a-d;b-a;c-b;d-e;e-c;

a) Selectively permeable membrane - d) Plasma membrane
Selectively permeable membrane is a characteristic of the plasma membrane, which is the outer layer of all cells. The plasma membrane controls the movement of materials into and out of the cell.
b) Traps solar energy - a) Chloroplast
Traps solar energy is a function of the chloroplasts, which are organelles found in plant cells. Chloroplasts use sunlight to convert carbon dioxide and water into glucose, which is the plant's food source.
c) Cyanobacteria - b) Prokaryotic
Cyanobacteria are a type of prokaryote, which means they do not have a nucleus. Cyanobacteria are photosynthetic, meaning they can trap solar energy and use it to produce food.
d) Detoxification - e) SER
Detoxification is the process of removing harmful substances from the body. SER (Smooth Endoplasmic Reticulum) is involved in detoxification by breaking down harmful substances.
e) Endoplasmic reticulum – c) Transport of minerals
By promoting the transfer of materials from its own membrane to other cellular sites, the smooth endoplasmic reticulum contributes to intracellular transport. To guarantee appropriate sorting and transport of lipids and proteins to their ultimate locations inside the cell, it can also interact with other organelles, such as the Golgi apparatus.

23. Correct option: (C) Leucoplasts

Plant cells have plastids as organelles. They are in charge of the manufacture and storage of numerous compounds. Chloroplasts, chromoplasts, and leucoplasts are the three primary categories of plastids.
A kind of plastids known as leucoplasts lacks pigment and is consequently colorless. They have a major role in the storage of proteins, lipids, and starches. Leucoplasts are widespread in the non-photosynthetic tissues of plants, including the roots, tubers, and seeds, where they are essential for storing and supplying the plant with nutrients.
The green plastids called chloroplasts are in charge of photosynthesis. Chlorophyll, a green pigment found in them, absorbs sunlight and uses it to change water and carbon dioxide into glucose. Chromoplasts are the colored plastids that give flowers, fruits, and other plant components their distinctive hues.
Therefore, the only colorless plastids are leucoplasts. So, the answer is C.

24. Correct option: (D) Chloroplast to chromoplasts

Chloroplasts, the organelles in unripe green fruits that are essential for photosynthesis, are present. Chloroplasts are made of the green pigment chlorophyll, which utilizes sunlight to absorb carbon dioxide and water and turn them into glucose.

The green pigment chlorophyll is destroyed when a fruit ripens, and the chloroplasts start to disintegrate. This makes the carotenoids and anthocyanins, two additional fruit pigments, more noticeable. These pigments are what give mature fruits their distinctive colors, including red, yellow, orange, and purple.

Thus, the conversion of the fruit's chloroplasts to chromoplasts accounts for how an unripe, green fruit changes color as it ripens. A particular class of plastid that has carotenoid pigments called chromoplasts. The deterioration of the chloroplasts and the buildup of carotenoid pigments are responsible for the fruit's color shift.

25. Correct option: (B) Plasmolysis

In a hypertonic solution, plasmolysis is the contraction of the cytoplasm away from the cell wall. This happens because osmosis—the flow of water from an area of high-water concentration to an area of low water concentration—causes the water inside the cell to escape the cell.

A solution that contains more solutes than the cell has a hypertonic solution. This indicates that there is less water present in the cell than there is in the solution. Osmosis causes water to leave the cell as a result, which causes the cytoplasm to contract.

The plasma membrane is detached from the cell wall during frontolysis. The destruction of the cell wall by acids is known as acidolysis. The disintegration of a cell by enzymes is known as allolysis.

26. Correct option: (B) They help with the process of cell division

Organelles that are membrane-bound and present in plant cells are called vacuoles. The vacuole serves several other essential functions in plant cells. These include storing toxic metabolic wastes (option A), helping to maintain turgidity (option C), and providing structural support (option D). The vacuole is a large, fluid-filled organelle in plant cells that plays a crucial role in maintaining the cell's shape, turgor pressure, and storing various substances.

Vacuoles don't participate in the division of cells, nevertheless. A cell can split into two new cells through a process called cell division. Many different genes regulate this intricate process. This procedure excludes the use of vacuoles.

27. Correct option: (C) Endoplasmic reticulum

All cells include the endoplasmic reticulum (ER), which is a network of membranes.

Rough ER, which has ribosomes and synthesizes proteins, and smooth ER, which lacks ribosomes and is involved in the synthesis of lipids, are the two different forms of ER.

The Golgi apparatus is where the ER-produced proteins and lipids are processed, packed, and delivered after being carried there.

Organelles called lysosomes are in charge of breaking down cellular trash. Chromosomes are biological entities that house a cell's genetic material. Organelles called mitochondria are in charge of a cell's ability to produce energy.

28. Correct option: (C) Osmosis

Water travels across a semi-permeable membrane known as an osmotic membrane by the process of osmosis from an area of greater water concentration (lower solute concentration) to a region of lower water concentration (higher solute concentration). Until the water concentration on both sides of the membrane is equal or balanced, this water flow continues. It is essential for many biological activities and is crucial for preserving the water balance in cells.

Options A and B are incorrect because diffusion is the movement of particles, including solute molecules, from a region of high concentration to an area of low concentration, whereas evaporation is the process of converting liquid into vapor.

29. Correct option: (A) Robert Hooke

The word "cell" was first used by English scientist Robert Hooke in 1665. He observed what he characterized as "little boxes" when examining a thin slice of cork under a microscope. He referred to these containers as "cells" since they reminded him of the honeycomb's cells. The finding made by Hooke marked a significant advance in our knowledge of life. It demonstrated how cells are the fundamental building block of life and how they are present in all living things.

The first person to use a microscope to view cells was Anton van Leeuwenhoek. He did not, however, invent the word "cell".

Himmler was a top Nazi official who was in charge of the Holocaust. From 1985 until 1991, Gorbachyey served as the Soviet Union's president. He had nothing to do with the discovery of cells, and they weren't scientists.

30. Correct option: (D) ATP

Adenosine triphosphate, or ATP, is frequently described as the "energy currency" of the cell. Within cells, it is a molecule that stores and transmits energy. The breakdown of glucose and other energy-dense compounds occurs during cellular respiration, notably in the mitochondria. This process results in the production of ATP. One of ATP's high-energy phosphate bonds can be broken by cells to swiftly release energy when they need it for different tasks.

This energy fuels cellular activity. Endoplasmic reticulum (Option A) and oxygen (Option B) are incorrect since they are not directly linked to being the energy unit of the cell. Option C (Mitochondria) is partially correct in the sense that mitochondria are the sites where ATP is produced, but ATP itself is the actual energy carrier in the cell.

LEVEL - II

1. Correct option: (A) Anton Van Leeuwenhoek

When Anton Van Leeuwenhoek first observed a cell, he said it was alive. Dutch scientist Anton Van Leeuwenhoek is frequently referred to be the founding father of microbiology. He made important contributions to microscopy in the 17th century and created powerful lenses that made it possible for him to examine microbes for the first time. He discovered little, single-celled creatures in 1674 that he referred to as "animalcules" or "little animals." He recognized that these bacteria were living things despite not completely understanding their nature and meticulously recorded his findings.

Robert Hooke first used the word "cell" to refer to the tiny spaces he saw in cork tissue. Schleiden suggested that all plants are composed of cells, whereas Robert Brown argued that plant cells have nuclei. They both missed seeing living cells, though. So, Anton Van Leeuwenhoek first saw a cell and described it as a live cell.

2. Correct option: (C) Cytology

Studying the structure, behavior, and functions of cells is known as cytology. Kytos, which means "hollow vessel," and logos, which means "study of," are the roots of the term "cytology." It is a field of biology that focuses on comprehending the cell, the building block of life. Cytologists inspect and thoroughly examine cells using a variety of methods and equipment, including microscopes.

Cardiology is the study of the heart. The heart is an organ, not a cell. Cosmology is the study of the universe. The universe is made up of many cells, but it is not itself a cell.

Histology is the study of tissues. Tissues are made up of cells, but they are not themselves cells.

Cytologists examine cells, which helps us comprehend the intricacies of living things and how they work at the microscopic level. So, the study of cell is known as Cytology.

3. Correct option: (D) All the above

Three scientists, Matthias Schleiden, Theodore Schwann, and Rudolf Virchow, worked together to develop the hypothesis of the cell.

German botanist Matthias Schleiden investigated plant tissues and came to the conclusion that all plants are made up of cells. German naturalist Theodore Schwann expanded on this notion and claimed that all animal tissues likewise consist of cells. The first two elements of the cell theory were developed jointly by Schleiden and Schwann.

German physician Rudolf Virchow believed that all cells develop from pre-existing cells.

Thus, the three scientists who worked together to develop the cell hypothesis were important in its development.

4. Correct option: (D) Cell

The smallest unit of life is a cell. All living things, from small unicellular creatures like bacteria to sophisticated multicellular organisms like plants and animals, including humans, are made up of these fundamental building components. Cells perform vital tasks required for the survival and operation of living creatures.

Each cell is a standalone organism with a unique combination of structures and organelles that collaborate to support life activities. Organs are made up of different types of tissues functioning together, tissues are grouped together to create tissues, and organ systems are made up of multiple organs working together to carry out particular body duties.

5. Correct option: (B) Ostrich

The biggest known single cell in the animal kingdom is the egg of an ostrich. Ostrich eggs may weigh up to 1.4 kilograms (3 pounds), which is quite a sizeable egg. These huge eggs are designed to support the weight of the incubating parent while providing the developing ostrich embryo with the nourishment and security it needs until it hatches. Native to Africa, ostriches are flightless birds that lay and nurture their eggs on the ground.

The other answer choices are incorrect. The eggs of elephants, camels, and cows are all much smaller than the ostrich egg.

6. Correct option: (A) RBC

In the human body, red blood cells (RBCs) are the most prevalent type of cell. They are in charge of delivering oxygen to the tissues and eliminating carbon dioxide. RBCs have a diameter of around 7.0 μm, which is comparable to a bacterium. They can effectively carry out their job in the bloodstream because to their small size and distinctive biconcave shape, which maximizes the surface area available for gas exchange.

The blood also contains white blood cells (WBCs); however, they are considerably smaller than RBCs. WBCs are in charge of preventing infection. A WBC's diameter typically ranges between 10 and 15 μm.

RBC is the only option that is accurate as a result.

7. Correct option: (D) None

The correct answer is not provided in the options. The cell membrane, also known as the plasma membrane, is the outermost membrane of the cell. It serves as a semi-permeable barrier around the cell and regulates how things enter and exit the cell.

A phospholipid bilayer, or two layers of lipids, composes the plasma membrane. The hydrophilic (water-loving) heads of the lipids face the inside and outside of the cell, whereas the hydrophobic (water-fearing) tails of the lipids face each other.

The jelly-like material that fills a cell is called cytoplasm. It includes all of the organelles, which are the minute structures that carry out the operations of the cell. The control center of the cell is the nucleus. It has the DNA of the cell, which is the genetic material responsible for a cell's structure and function. Proteins and other chemicals are transported throughout the cell via the endoplasmic reticulum, a network of membranes. So, the outer most membrane of the cell is the plasma membrane, and none of the other answer choices are correct.

8. Correct option: (D) All the above

The plasma membrane is a very important part of the cell. It has a number of functions, including:

Providing structure and shape to the cell. The plasma membrane is a thin, flexible barrier that surrounds the cell and gives it its shape.

- Acting as a mechanical barrier for protection. The plasma membrane helps to protect the cell from its environment. It prevents harmful substances from entering the cell and keeps the cell's contents from leaking out.
- Regulating the movement of substances across the membrane. The plasma membrane controls what substances can enter and leave the cell. This is important for maintaining the cell's internal environment.

So, option D is the correct answer as it includes all the functions of the plasma membrane.

9. Correct option: (B) Hypertonic

The concentration of solutes outside a cell is higher than inside the cell when the cell is submerged in a hypertonic solution. As a result, water osmotically diffuses from the cell into the surrounding solution, shrinking the cell. In an effort to balance the concentrations on both sides of the cell membrane, water moves from a region of lower solute concentration (inside the cell) to an area of greater solute concentration (outside the cell).

A cell will enlarge when submerged in a hypotonic solution because water will enter the cell through osmosis. This is true because water always moves from a region with low solute concentration to one with high solute concentration.

A solution that contains the same amount of solute as the cell is said to be isotonic. When a cell is submerged in an isotonic solution, there is no net water flow into or out of the cell.

So, the answer to the question is Hypertonic.

10. Correct option: (A) Exocytosis

Exocytosis is the process through which substances, such as waste products or proteins, are moved from inside the cell to the outside. Exocytosis is the discharge of components from vesicles into the extracellular space after they have fused with the cell membrane. Numerous cellular processes, such as the release of hormones, neurotransmitters, and other cellular byproducts, depend on this mechanism.

The process by which materials travel from the exterior of the cell to the interior is known as endocytosis. It is thus a form of active transportation. The process of osmosis involves the transfer of water molecules from a region with low solute concentration to one with high solute concentration.

So, the answer to the question is Exocytosis.

11. Correct option: (B) Endocytosis

The process of endocytosis is how cells absorb substances from their surroundings. Because it is a form of active transport, the components are transported inside the cell using energy. Phagocytosis and pinocytosis are the two forms of endocytosis.

Phagocytosis is the procedure through which cells take in big objects like bacteria. The particle is enclosed by the cell membrane, which creates a vacuole around it. The particle is broken down by the enzymes, and the cell then takes the nutrients in.

Exocytosis is the process by which cells discharge items to the surrounding environment from inside. Vesicles, which contain the materials and fuse with the cell membrane to release their contents, are used to package the materials.

So, the process by which materials move into the cell by engulfing of the food is called endocytosis. The answer is B.

12. Correct option: (A) Plant

The plasma membrane of plant cells is protected on the outside by the cell wall.

Plants have a protective, hard structure called a cell wall that is mostly made of polysaccharides like cellulose and hemicellulose. It gives the plant cell structural support and defense, assisting in maintaining its form and preventing it from bursting or collapsing under various circumstances.

Animal cells, however, do not have a cell wall; instead, they are covered and protected by a flexible plasma membrane.

13. Correct option: (A) Robert brown

Robert Brown, a Scottish botanist, made the discovery that cells had nuclei in 1831. He was using a microscope to examine orchid cells when he noticed a tiny, dark-staining body in the center of the cell. He referred to it as the nucleus.

Brown's finding was significant since it demonstrated that the nucleus was a separate cell component and not merely a collection of proteins. It is now understood that the nucleus, which houses the cell's DNA, serves as its administrative hub.

An English scientist named Robert Hooke is credited with discovering cells in 1665. He failed to find the nucleus, though.

Robert Brown was the scientist who gave the nucleus the name "Nucleus" in 1831.

14. Correct option: (C) Two

Rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER) are two separate compartments within the endoplasmic reticulum (ER).

Ribosomes, the organelles in charge of protein synthesis, are embedded in the rough endoplasmic reticulum (RER). Proteins that will be exported from the cell are created with the help of the RER.

The ribosome-free smooth endoplasmic reticulum (SER) is present. The SER is engaged in a number of processes, including as calcium ion storage, lipid synthesis, and the detoxification of poisonous chemicals.

Hence, Endoplasmic reticulum is divided into C) Two distinct compartments.

15. Correct option: (D) Ribosomes

The endoplasmic reticulum's (ER) rough surface is caused by the presence of ribosomes on its cytoplasmic surface. Ribosomal RNA (rRNA) and proteins make up the small, intricate structures known as ribosomes. They are in charge of the cell's protein production.

The endoplasmic reticulum, also known as the "rough endoplasmic reticulum" or RER, has a rough look when ribosomes are connected to the cytoplasmic side of the ER.

The RER is involved in the synthesis of proteins that are destined for secretion from the cell, incorporation into the cell membrane, or transport to other organelles.

16. Correct option: (A) Proteins

In the cell, proteins are made and transported through the Rough Endoplasmic Reticulum (RER). It is a membrane-bound organelle that has a cytoplasmic surface covered in ribosomes.

The ribosomes on the RER synthesize the proteins straight into the lumen (interior) of the RER when a cell wants to manufacture proteins. In order to achieve their functional structure, the freshly synthesized proteins are next processed, folded, and changed in the RER. The RER assists in moving proteins to their ultimate locations inside the cell or for secretion outside the cell once they have been appropriately digested.

The Rough Endoplasmic Reticulum does not directly synthesize or transport carbohydrates. Depending on the individual kind of carbohydrate, they are largely synthesized in the cytoplasm or other organelles such as the smooth endoplasmic reticulum and chloroplasts (in plant cells).

17. Correct option: (D) Mitochondria

The crucial function they play in generating energy has led to the nickname "powerhouse of the cell" for mitochondria. Eukaryotic cells include these organelles, which are in charge of producing adenosine triphosphate (ATP) via cellular respiration. The energy-storing and energy-transfer molecule ATP gives the fuel for many cellular operations by storing and transferring energy inside the cell.

Golgi complexes are crucial in the packaging and processing of proteins, but they don't make ATP. Water, nutrients, and waste materials are all stored in vacuoles, which are storage organelles. Plastids are exclusively present in plant cells and are essential for photosynthesis. Thus, mitochondria are the organelle referred to as the powerhouse of the cell.

18. Correct option: (D) None

In the cytoplasm of the cell, amino acids and fatty acids are created. The cytoplasm, which makes up the majority of a cell's volume but excludes the nucleus, is a jelly-like material that houses all of the organelles.

Although they are important in the packaging and processing of proteins, Golgi bodies do not produce fatty or amino acids. Water, nutrients, and waste materials are stored in vacuoles, which are storage organelles. Only present in plant cells, plastids play a role in photosynthesis.

As a result, none of the organelles mentioned above are involved in the production of amino acids or fatty acids.

19. Correct option: (B) Proteins

Protein synthesis occurs at ribosomes. All cells, both plant and animal, have them in the cytoplasm. Two subunits, a big subunit and a small subunit, make up ribosomes. A functioning ribosome is created by the union of the two subunits.

The two steps involved in protein synthesis. First, messenger RNA (mRNA) is synthesized from DNA. The genetic code for a protein is contained in a molecule called mRNA. The mRNA is translated into a protein at the ribosome in the second step.

In order to put together the amino acids into a protein, the ribosome reads the mRNA codons. Peptide bonds are used to connect the amino acids. After that, the ribosome releases the finished protein into the cytoplasm.

So, the organelles that are the site of protein synthesis are ribosomes. The answer is B.

20. Correct option: (C) Lysosomes

Lysosomes are referred to as "suicidal bags of the cell" because they contain hydrolytic enzymes that can disassemble biological components. Lysosomes contain various hydrolytic enzymes that are capable of breaking down and digesting cellular waste, foreign materials, and even damaged organelles. These enzymes have the power to harm

or kill a cell if a lysosome ruptures. This is the reason why lysosomes are frequently referred to as "suicide bags."

Because they lack hydrolytic enzymes, ribosomes are not referred to as "suicidal bags." Protein synthesis is carried out by ribosomes, which are organelles.

Rough endoplasmic reticulum has the initials RER. Ribosomes cover a specific kind of endoplasmic reticulum in this cell.

So, the organelles that are called as "suicidal bags of the cell" are lysosomes. The answer is C.

21. Correct option: (B) Plant

Plant cells, some varieties of algae, and protists have plastids as organelles. Animal cells don't contain them. Plastids are necessary for many cellular functions, such as photosynthesis, the storage of starch and pigments, and the production of lipids and amino acids.

Chlorophyll, a green pigment that absorbs light energy during photosynthesis, is a component of chloroplasts, a particular form of plastid. Leucoplasts, which are involved in the manufacture and storage of numerous compounds, such as starch and oils, and chromoplasts, which store the pigments that give fruits and flowers their vivid colors, are other types of plastids.

As opposed to plants, which undertake photosynthesis and have diverse organelles to suit their metabolic demands, animal cells do not have plastids or chloroplasts.

So, plastids are present only in plant cells. The answer is B.

22. Correct option: (A) Cytosomes

In 1965, Christian de Duve used the word "cytosome" to refer to these organelles. Because they are located in the cytoplasm, he picked this nomenclature.

Cytosomes are another name for microbodies. The cytoplasm of eukaryotic cells contains these tiny, membrane-bound organelles. They take part in several metabolic processes, such as the oxidation of lipids, the creation of hydrogen peroxide, and the detoxification of poisonous chemicals. The two most well-known varieties of microbodies are peroxisomes and glyoxysomes.

23. Correct option: (D) Centrioles

Animal cells have structures called centrosomes that house two centrioles, which are cylindrical organelles. These granule-like centrioles are essential for mitosis and meiosis, the processes of cell division. The centrioles travel to the cell's opposing ends during cell division and assist in the creation of the mitotic spindle, which is necessary for the appropriate distribution of chromosomes to the daughter cells.

A matrix of proteins and other molecules that play a role in organizing the cell's microtubules may also be found in the centrosome in addition to centrioles. Not all cell types include centrioles, and some eukaryotic cells do not. Despite the absence of centrioles in plant cells, the microtubule-organizing center (MTOC) in these cells serves a similar purpose.

24. Correct option: (A) Rudolf Von Virchow

A German physician and pathologist named Rudolf Virchow is credited with creating cellular pathology. He studied animal cell division in 1855 and put out the idea that all cells are descended from earlier cells. The term for this idea is the "omnis cellula e cellula principle," which is derived from the Latin phrase "every cell comes from a cell."

An English scientist named Robert Hooke is credited with discovering cells in 1665. He did not, however, see cell division.

Robert Brown, a Scottish botanist, is credited with discovering cells' nuclei in 1831. Additionally, he missed seeing cell division.

Rudolf Von Virchow, then, was the scientist who first noticed cell division in 1855. The response is A.

25. Correct option: (A) Prokaryotic

Prokaryotic cells are those that do not contain a nucleus. Bacteria and archaea are home to the most basic sort of cell, the prokaryotic cell. The nucleus, mitochondria, and chloroplasts are examples of organelles that are membrane-bound yet absent in them. The cytoplasm, the jelly-like material that makes up the inside of prokaryotic cells, is where the DNA is located.

A nucleus and other membrane-bound organelles are present in eukaryotic cells, on the other hand. Each and every other type of life, such as plants, mammals, fungi, and protists, has eukaryotic cells.

So, the cell without nucleus is known as prokaryotic cell. The answer is A.

3. Why do we fall ill?

Solutions

LEVEL-I

1. Correct option: (D) Elephantiasis

Lymphatic filariasis, commonly known as elephantiasis, is a painful and profoundly disfiguring disease. Elephantiasis can last for a very long time and has long-term effects on the body. Therefore, it is a chronic disease.

2. Correct option: (C) Jaundice

Jaundice happens when too much bilirubin is made in the blood or when the liver is damaged and cannot get rid of bilirubin from the body. Other than yellow skin, people with jaundice may have stomach pain, nausea, fever, weight loss, or itching.

3. Correct option: (A) Microbes

nfectious diseases are disorders caused by organisms — such as bacteria, viruses, fungi or parasites. Many organisms live in and on our bodies. They're normally harmless or even helpful. But under certain conditions, some organisms may cause disease.

4. Correct option: (C) Cancer

Mutations or genetic changes on DNA results in changing genetic makeup of a cell resulting in altering the behaviour and metabolism of cells which in return cause cells to divide indefinitely.

5. Correct option: (C) Robin Warren

Barry J. Marshall and Robin Warren, two Australian researchers who discovered the bacterium Helicobacter pylori and deciphered its role in gastritis and peptic ulcer disease, have been awarded this year's Nobel Prize in Physiology or Medicine.

6. Correct option: (B) 50

Upon analysis of negative-stained SARS-CoV-2 articles by electron microscopy, different researchers have had varying results, but the diameter of the virus has been found to range between 50 nm to 140 nm.

7. Correct option: (A) Small intestine

Roundworms are small organisms that can live in your intestine, which is part of your digestive system. Roundworms can live in the human small intestine for a long time and can cause infections. Ascaris is an intestinal parasite of humans. It is the most common human worm infection. The larvae and adult worms live in the small intestine and can cause intestinal disease.

8. Correct option: (A) bacteria

There are many organisms that can cause skin infections including bacteria, viruses and fungi. Bacteria like Staphylococcus aureus may cause skin infections, Fungi including Trichophytons cause ringworms, and viruses like Herpes cause cold sores. Most prominent infections are caused by bacteria.

9. Correct option: (B) tuberculosis

TB is spread through the air from one person to another. TB bacteria are passed through the air when someone who is sick with TB disease of the lungs or throat coughs, speaks, laughs, sings, or sneezes. Anyone near the sick person with TB disease can breathe TB bacteria into their lungs.

10. Correct option: (A) Mosquitoes

Most of the time mosquitoes feed on sugary substances like plant nectars. However, when it's time for female mosquitoes to lay eggs, they need the extra protein that comes from blood supplies. Only female mosquitoes bite living hosts such as humans and other warm-blooded animals.

11. Correct option: (B) AIDS

HIV (human immunodeficiency virus) is a virus that attacks the body's immune system. If HIV is not treated, it can lead to AIDS (acquired immunodeficiency syndrome). There is currently no effective cure. Once people get HIV, they have it for life. But with proper medical care, HIV can be controlled.

12. Correct option: (A, B, D) Small pox, Chicken pox and AIDS

Small pox, Chicken pox and AIDS are viral diseases caused by Variola virus, varicella zoster virus and HIV respectively. Tuberculosis is a bacterial infection caused by Mycobacterium tuberculosis.

13. Correct option: (D) Diabetes

Diabetes is one of the four major types of noncommunicable diseases (cardiovascular disease, diabetes, cancer and chronic respiratory diseases). It is a chronic condition that occurs when the body either does not produce enough insulin or cannot effectively use the insulin it does produce.

14. Correct option: (D) Female Anopheles mosquito

In the case of malaria, the vector is the female anopheline mosquito and the disease-causing organism is the malaria parasite. Humans and anopheline mosquitoes are both considered to be the parasite's hosts.

15. Correct option: (C) TB

BCG, or bacille Calmette-Guerin, is a vaccine for tuberculosis (TB) disease. Many foreign-born persons have been BCG-vaccinated. BCG is used in many countries with a high prevalence of TB to prevent childhood tuberculous meningitis and miliary disease.

16. Correct option: (A) Cholera

The cholera bacteria is passed through feces (poop). It is spread by eating or drinking food or water contaminated by the feces (poop) of an infected person. This occurs more often in underdeveloped countries lacking proper water supplies and sewage disposal.

17. Correct option: (D) Bed bugs

Plague is an infectious disease caused by the bacteria Yersinia pestis, a zoonotic bacterium, usually found in small mammals and their fleas. It is transmitted between animals through fleas. Humans can be infected through: the bite of infected vector fleas.

18. Correct option: (D) TB

Tuberculosis is a chronic, progressive mycobacterial infection, often with an asymptomatic latent period following initial infection. TB bacteria can live in the body without making you sick. This is called latent TB infection.

19. Correct option: (B) Alexander Flemming

In 1928 Dr Alexander Fleming returned from a holiday to find mould growing on a Petri dish of Staphylococcus bacteria. He noticed the mould seemed to be preventing the bacteria around it from growing. He soon identified that the mould produced a self-defence chemical that could kill bacteria and later on named it penicillin.

20. Correct option: (D) Cholera

Cholera is an acute diarrheal illness caused by infection of the intestine with Vibrio cholerae bacteria. People can get sick when they swallow food or water contaminated with cholera bacteria.

LEVEL-II

1. Correct option: (B) Increases

It can be hard to follow the work schedule, but the good thing is that you are at the comfort of your home, so utilize it at its best. Take a good 10-minute walk every 2 or 3 hours or do stretching and take a good diet and sanitary hygiene. This will not only increase your productivity but would also prevent you from diseases.

2. Correct option: (A) Good economic
Social harmony and good economic conditions are necessary for good health.
3. Correct option: (D) All the above
Community health is defined as the group of individuals living in a society who aims to maintain, protect and improve the health of the people. The whole community aims to maintain the health of people by providing safe drinking water, proper disposal of sewage, by providing proper medical facilities, and by providing proper health education to the people in the community.
4. Correct option: (C) Disturbed
The term "disease" is a condition in which the normal functioning of the body is disturbed. Many times, such conditions are not very serious and get cured, but sometimes they become serious and even fatal.
5. Correct option: (A) Short
Acute diseases come on rapidly, and are accompanied by distinct symptoms that require urgent or short-term care, and get better once they are treated.
6. Correct option: (D) Both A and B
Acute diseases last for a shorter period of time. They do not have any long-term effect on the patient. E.g., common cold, cough, etc.
7. Correct option: (B) Chronic
Reducing a person's ability to work efficiently and experiencing difficulties in learning at school are symptoms typically associated with chronic conditions or long-term issues rather than acute ones. They develop over time and can have a gradual onset. Symptoms like reduced work efficiency and learning difficulties can be linked to chronic conditions that affect a person's overall health, cognitive abilities, or emotional well-being over an extended period.
8. Correct option: (B) Chronic
Chronic diseases are typically long-lasting conditions that develop gradually and persist over time. Tuberculosis (TB) is caused by the bacterium Mycobacterium tuberculosis and primarily affects the lungs. It often presents with symptoms such as cough, weight loss, fatigue, and fever. If left untreated, TB can persist for an extended period, making it a classic example of a chronic disease.
9. Correct option: (B) Pathogens
The disease-causing organisms like bacteria, viruses, fungi, protozoans, worms, etc. are referred to as pathogens.
10. Correct option: (D) virus
Chickenpox is caused by the varicella-zoster virus (VZV). This virus is responsible for the development of chickenpox (varicella) during the initial infection and can later reactivate to cause shingles (herpes zoster) in some individuals.
11. Correct option: (B, D) Typhoid and Cholera
Cholera is caused by the bacterium Vibrio cholerae and primarily affects the digestive system, leading to severe diarrhea and dehydration. Typhoid fever is indeed caused by the bacterium Salmonella typhi. It's a bacterial infection that can lead to high fever, abdominal pain, and other symptoms.
12. Correct option: (D) None of the above
Leprosy, Syphilis, and TB are all bacterial diseases caused by Mycobacterium leprae, Treponema pallidum and Mycobacterium tuberculosis respectively.
13. Correct option: (B) Unhealthy
A nutritious diet provides essential nutrients that the body needs to function properly, maintain good health, and prevent various health issues. When a person lacks proper nutrition due to an inadequate diet, their body may not receive the necessary vitamins, minerals, proteins, and other nutrients it requires.

14. Correct option: (A) Congenital
Congenital refers to a condition or trait that is present at or from birth. It can refer to both genetic conditions and conditions that might have been influenced by prenatal factors.
15. Correct option: (B) Acquired
Acquired diseases are those that are not present at birth and are typically developed or contracted due to various factors such as infections, lifestyle choices, environmental influences, and so on. These diseases are not inherited and do not pass from one generation to another through genetic means.
16. Correct option: (A) Communicable
A communicable disease (also known as an infectious or contagious disease) is a type of disease that can be transmitted from one person, animal, or object to another through various means. This can include direct contact, respiratory droplets, contaminated food or water, vectors like insects, and more. Some examples of communicable diseases include the common cold, influenza, tuberculosis, and COVID-19.
17. Correct option: (B) Infectious
Tuberculosis (TB) is an infectious disease that most often affects the lungs and is caused by a type of bacteria. It spreads through the air when infected people cough, sneeze or spit.
18. Correct option: (B) Non-infectious
Scurvy is a disease resulting from a lack of vitamin C (ascorbic acid). Early symptoms of deficiency include weakness, fatigue, and sore arms and legs. Thus, it is non-infectious.
19. Correct option: (C) Common cold
Colds are primarily transmitted from person to person via hands that are contaminated with nasal secretions. Less often, viruses that cause colds can be transmitted by touching a contaminated surface or via sneezing or coughing. Direct contact — People with colds typically carry the cold virus on their hands
20. Correct option: (C) Antibiotic
Penicillin is an antibiotic that is it is used to treat bacterial infections. It was discovered by Sir Alexander Fleming in the year 1920.
21. Correct option: (D) None
AIDS is a viral disease caused by HIV virus through blood transfusion via infected needles, sexual contact, and mother to infant through breast feeding. It is present in body fluids.
22. Correct option: (A) Water borne
Drinking water should be filtered and boiled, ozonised, and treated with chlorine before making it potable, thereby avoiding water borne infections. This ensures that no pathogenic bacteria like typhus and cholera are persisting there.
23. Correct option: (A) transmissible
Communicable diseases, also known as infectious diseases or transmissible diseases, are illnesses that result from the infection, presence and growth of pathogenic (capable of causing disease) biologic agents in an individual human or other animal host.
24. Correct option: (C) Are vectors for many diseases
Mosquitoes are known to transmit various diseases such as malaria, dengue fever, Zika virus, and more. Preventing mosquito breeding in our surroundings is crucial to reduce the spread of these diseases.
25. Correct option: (B) Tuberculosis
BCG, or bacille Calmette-Guerin, is a vaccine for tuberculosis (TB) disease. Many foreign-born persons have been BCG-vaccinated. BCG is used in many countries with a high prevalence of TB to prevent childhood tuberculous meningitis and miliary disease.

4. Improvement in Foor Resources

Solutions

LEVEL-I

1. Correct option: (D) Phosphorous

Phosphorus is one of the three macronutrients essential for plant growth. It is required for the photosynthesis process, converting the sun's energy into food for the plant. It is also required for strong root development. A plant must be able to access phosphorus to ensure a healthy growing cycle.

2. Correct option: (D) Phosphorous

Phosphorous is used by plants to help form new roots, make seeds, fruit and flowers. It's also used by plants to help fight disease.

3. Correct option: (B) Black Gram

Black gram can be grown on variety of soils ranging from sandy soils to heavy cotton soils. The most ideal soil is a well-drained loam with pH of 6.5 to 7.8. Black gram cannot be grown on alkaline and saline soils. Land is prepared like any other kharif season pulse crop.

4. Correct option: (D) All of these

Intercropping is the practice of growing two or more crops in proximity. For instance, the spaces between the rows of the Jackfruit trees should be cultivated and used for the production of annual or perennial intercrops or both. Examples of such intercrops are Citrus, Banana, Pineapple, Corn, Peanut (Groundnut), and other Pulses, Spices, and vegetables.

5. Correct option: (D) Both A & B

Root nodules are mostly present in the leguminous plants. E.g. Pulses, beans, soybeans, etc. These nodules provide shelter to certain bacteria which aids in nitrogen fixation.

6. Correct option: (A) Rhizobia

Rhizobium is a soil bacteria that fixes nitrogen belonging to the genus of gram-negative bacteria. Rhizobium species from endosymbiotic nitrogen-fixing associations with legumes. The bacteria colonize plant cells in the roots, where they use the nitrogenase enzyme to convert atmospheric nitrogen to ammonia and then provide organic nitrogen compounds such as glutamine.

7. Correct option: (D) Both A & B

Manure is an organic substance obtained by the decomposition of animal and plant wastes. It helps to maintain the fertility of the soil without having deleterious effects on soil as well as the crops grown on it.

8. Correct option: (D) All of these

These concentrated manures are produced using basic materials that are either plant- or animal-derived. Oil cakes, bone meal, fish meal, blood meal, and meat meal are some of the concentrated organic manures that are frequently employed. These include Jatropha, Neem and Groundnuts.

9. Correct option: (D) Both A and C

Natural organic manures are generally divided into two types. One is concentrated organic manures and the other is macro-organic manure. Animal excreta, compost, deep litter are the examples of macro-organic manure.

10. Correct option: (C) Insecticides

Insecticides are pesticides that are formulated to kill, harm, repel or mitigate one or more species of insect. Insecticides work in different ways. Some insecticides disrupt the nervous

system, whereas others may damage their exoskeletons, repel them or control them by some other means.

11. Correct option: (D) All of these

The three options that have been used for increasing food production are (i) agro-chemical based agriculture (ii) organic agriculture; and (iii) genetically engineered crop-based agriculture.

12. Correct option: (B) Carbohydrate

Starch is a carbohydrate commonly found in nature and one of the primary sources of food energy for human beings. It is regularly eaten in the form of wheat, rice, potatoes, and other staple foods cultivated throughout the world. Alongside fiber and sugar, starch is one of the three main categories of carbohydrates.

13. Correct option: (C) Drip irrigation

Drip irrigation saves wastage of water because it supplies water directly to the roots.

14. Correct option: (B) Nitrogen

The primary macronutrients are Nitrogen (N), Phosphorus (P), and Potassium (K). Nitrogen is essential for plant development, since it plays a fundamental role in energy metabolism and protein synthesis. Nitrogen is absorbed by the plant in the form of a nitrate. This macronutrient is directly related to plant growth.

15. Correct option: (D) All of these

Fertilisers and manures contain plant nutrients such as nitrogen, phosphorous and potassium, etc. So, when fertilisers and manures are added to the soil in the fields, then the soil gets enriched with nutrients like nitrogen, phosphorous and potassium, etc.

Nutrients are replenished by certain different microorganisms that are found in the soil or also by the death of certain bacteria such as the rhizobium.

16. Correct option: (B) 50 to 150 kg

Values estimated for various legume crops and pasture species are often impressive, commonly falling in the range of 50 to 150 kg of N ha^{-1} $year^{-1}$.

17. Correct option: (D) All the above

Panchagavya or panchakavyam is a mixture used in traditional Hindu rituals that is prepared by mixing five ingredients. The three direct constituents are cow dung, urine, and milk; the two derived products are curd and ghee. These are mixed in proper ratio and then allowed to ferment.

18. Correct option: (B) 46%

The percentage of nitrogen present in urea is 46.6%. Urea contains four elements: carbon, hydrogen, nitrogen, and oxygen. The molecular formula for urea is CH4 N2 O. This compound is made up of a carbonyl (C=O) group attached to two nitrogen groups (NH2). A carbonyl carbon is one that is double bonded to oxygen.

19. Correct option: (A) Tungrovirus on paddy

Tungro virus disease is transmitted by leafhoppers, wherein the most efficient vector is the green leafhopper.

20. Correct option: (A) Rhizobium

Rhizobium is a soil bacteria that fixes nitrogen belonging to the genus of gram-negative bacteria. Rhizobium species from endosymbiotic nitrogen-fixing associations with legumes. The bacteria colonize plant cells in the roots, where they use the nitrogenase enzyme to convert atmospheric nitrogen to ammonia and then provide organic nitrogen compounds such as glutamine.

21. Correct Matching is: a-T, b-S, c-P, d-R, e-Q

(a) Kharif crop – (T) Agricultural crops that arise in the rainy season

In India, the season is popularly considered to start in June and end in October. Kharif crops are usually sown at the beginning of the first rains during the advent of the south-west monsoon season, and they are harvested at the end of monsoon season (October–November).
(b) Rabi crop – (S) Agricultural crops that raise in winter
Rabi Crops are harvested in the spring season while it is sown in winter. The rabi crops are sown around mid-November, preferably after the monsoon rains are over, and harvesting begins in April / May
(c) Mixed crop – (P) Cultivation of more than one crop in the same field
In agriculture, multiple cropping or multicropping is the practice of growing two or more crops in the same piece of land during one year, instead of just one crop. When multiple crops are grown simultaneously, this is also known as intercropping.
(d) Weedicides – (R) Chemicals used to kill weeds
Weedicides are the chemicals which are generally sprayed all around the field to prevent the growth of weeds. An example of common weedicide is 2,4- D (2,4 dichlorophenoxy acetic acid) which acts as a dicot weedicide.
(e) Herbicides – (Q) Chemicals used to kill herbs
Herbicide stops the growth of plants whereas weedicides skill weeds. They are designed to kill specific types of pests and are used mainly on plants.

22. Correct option: a-S, b-P, c-T, d-Q, e-R

(a) Earthworm – (S) Vermicompost
Vermicomposting is a natural process whereby earthworms convert waste material with rigid structures into compost. The compost produced in this green process is traditionally and popularly used as a natural fertilizer for enhancing plant growth.
(b) Urine of cow – (P) Panchagavya
It consists of nine products viz. cow dung, cow urine, milk, curd, jaggery, ghee, banana, Tender coconut and water. When suitably mixed and used, these have miraculous effects.
(c) Gingelly – (T) Concentrated organic manure
(d) Zinc – (Q) Micronutrient
Zinc (Zn) is an important micronutrient for plants since it is involved in many key cellular functions such as metabolic and physiological processes, enzyme activation, and ion homeostasis.
(e) Phosphorous – (R) Macronutrient
The primary macronutrients are Nitrogen (N), Phosphorus (P), and Potassium (K). Nitrogen is essential for plant development, since it plays a fundamental role in energy metabolism and protein synthesis. Nitrogen is absorbed by the plant in the form of a nitrate. This macronutrient is directly related to plant growth

LEVEL-II

1. Correct option: (C) Both A and B

Rice has approximately 2.69% of protein, and they have high carbohydrates. At the same time, millets are low in carbohydrates than rice and contain a higher amount of protein.

2. Correct option: (B) Green gram

The mung bean or Organic green gram is a legume family plant. Its super high protein content has made it a popular ingredient for making plant-based alternatives to animal-based food items like eggs.

3. Correct option: (A) Rainy

In India, the season is popularly considered to start in June and end in October. Kharif crops are usually sown at the beginning of the first rains during the advent of the south-west monsoon season, and they are harvested at the end of monsoon season (October–November).

4. Correct option: (C) Winter

Rabi Crops are harvested in the spring season while it is sown in winter. The rabi crops are sown around mid-November, preferably after the monsoon rains are over, and harvesting begins in April / May.

5. Correct option: (C) Peas

Examples of Kharif crops

Paddy, maize, bajra, jowar, soybean, castor, cotton, sugarcane, turmeric, chilly, bitter gourd, guar, okra, brinjal,turmeric, ragi.

6. Correct option: (B) Hybridization

Hybridization refers to crossing between genetically dissimilar plants. This crossing may be intervarietal (between different varieties), interspecific (between two different species of the same genus) or intergeneric (between different genera).

7. Correct option: D) All the above

Cultivation practices and crop yield are related to weather, soil quality and availability of water. Since weather conditions such as drought and flood situations are unpredictable, varieties that can be grown in diverse climatic conditions are useful.

8. Correct option: (C) Both A and B

Crop management is the set of agricultural practices performed to improve the growth, development and yield of crops. It begins with a seedbed preparation, sowing of seeds and crop maintenance; and ends with crop harvest, storage and marketing.

9. Correct option: (B) Manure

Manure is obtained when the animal and plant waste are dumped by farmers in an open pit and leave it for days. The waste starts to decompose by the action of certain microorganisms, this process is named decomposition. Finally, the decomposed organic matter becomes Manure.

10. Correct option: (D) Both A and B

Air supplies two nutrients to plants namely carbon and oxygen. Water supplies hydrogen and soil supplies thirteen nutrient elements to the plants. Among these thirteen nutrients, six are macronutrients and the other seven nutrients are micronutrients.

11. Correct option: (B) Soil

Plants derive the required nutrients from the air, water, and soil. There are sixteen nutrients essential for the growth of plants. Carbon and Oxygen are supplied by water. The remaining thirteen nutrients are supplied by soil.

12. Correct option: (A) Macronutrients

Nutrients which are needed in large quantities are called macro nutrients.

13. Correct option: (B) Calcium

These are the most important elements required for crops and examples include nitrogen (N), potassium (K), calcium (Ca), phosphorus (P), magnesium (Mg), sulphur (S), oxygen (O), carbon (C), and hydrogen (H). Calcium is an essential micronutrient in plants. It serves as a constituent of cell walls and membranes and thus contributes to the structure of cells and the upholding of physical barriers against pathogens.

14. Correct option: (B) Copper

Copper (Cu) is one of eight essential plant micronutrients. Copper is required for many enzymatic activities in plants and for chlorophyll and seed production. Deficiency of copper

can lead to increased susceptibility to diseases like ergot, which can cause significant yield loss in small grains.

15. Correct option: (B) Inorganic

It is commonly prepared by decomposing plant and food waste, recycling organic materials, and manure. The resulting mixture is rich in plant nutrients and beneficial organisms, such as bacteria, protozoa, nematodes, and fungi.

16. Correct option: (A) Vermicompost

Vermicomposting is a natural process whereby earthworms convert waste material with rigid structures into compost. The compost produced in this green process is traditionally and popularly used as a natural fertilizer for enhancing plant growth.

17. Correct option: (C) Fertilizers

Fertilizers supply nitrogen, phosphorus and potassium. They are used to ensure good vegetative growth (leaves, branches and flowers), giving rise to healthy plants. Fertilizers are a factor in the higher yields of high-cost farming.

18. Correct option: (D) All the above

Excess use of fertilizers causes environmental pollution as their residual and unused amounts will become pollutants for air, water, and soil. It also initiates the process of eutrophication in the water bodies, decreasing the soil quality by killing the soil microflora, etc.

19. Correct option: (C) 3

There are three types of wells: dug, driven, and drilled. Drilled are the most common today.

20. Correct option: (A) Mixed cropping

Growing two or more crops simultaneously on the same piece of land is known as mixed cropping. This lowers the risk and provides some protection if one of the crops fails. For example - Wheat with gram.

21. Correct option: (B) Inter- Cropping

The practice of growing two or more crops simultaneously in definite rows in the same field is called inter cropping.

22. Correct option: (B) Crop rotation

Growing different crops on a piece of land in pre-planned succession is known as crop rotation.

23. Correct option: (D) All of them

Weeds are unwanted plants which grow along with the crop plants in the cultivated field. Some common weeds in India are Amaranthus (Chaulai), Parthenium (Gajar ghas), Cyperus rotundus (Motha), Chenopodium (Bathua), Xanthium (Gokhroo). Weeds compete with the crop plants for nutrients, space, light and water.

24. Correct option: (A) Animal

Animal husbandry is a scientific management of livestock, breeding and rearing of farm animals.

25. Correct option: (D) Both A and C

Most of the Indian beekeepers engaged in beekeeping activity are primarily doing the extraction of raw honey from the bee colonies maintained by them. Very few beekeepers are extracting wax & pollen.

5. Life Process
Solutions
Digestion and Absorption

LEVEL-I

1. Correct Option: (C) iii & iv

Cuscuta, ticks, and leeches are parasites that feed by living on or inside another organism's body. Cuscuta species are plant parasites. Since they lack chlorophyll, they cannot use photosynthesis to produce their sustenance. To grow, they depend on other plants for nutrients, which weakens the host plant. Most leeches are sanguivorous, that is they feed as blood sucking parasites on preferred hosts.

2. Correct Option: (B) humidity

There are many factors that can alter the rate of photosynthesis. However, one that does not affect this process is Humidity. Studies are indicative of the fact that humidity plays an almost negligible role in the process of photosynthesis.

3. Correct Option: (B) Remove starch from leaves

The process is known as destarching. This is done so because the result of the experiment is not affected.

4. Correct Option: (A) Bile

Bile is a digestive juice that has no enzymes. Bile is an alkaline liquid that helps in emulsification of fat i.e., it helps in breakdown of large fat globules into smaller droplets that increases its surface area and accelerates digestion.

5. Correct Option: (A) The entire body surfaces

In single celled animals like amoeba, the food is taken up by the whole body. The process by which the single celled animals feed is known as Phagocytosis. In this process, the single celled organisms bring the food into the inner part of the cell for digestion.

6. Correct Option: (A) C.B. Van Neil

By studying purple Sulphur bacteria and green Sulphur bacteria he was the first scientist to demonstrate, in 1931, that photosynthesis is a light-dependent redox reaction in which hydrogen from an oxidizable compound reduces carbon dioxide to cellular materials.

7. Correct Option: (C) Starch

In the presence of starch, iodine changes its color from brown to blue-black or purple. So, iodine can be used to test the presence of starch.

8. Correct Option: (C) Joseph priestly

In 1770 he performed a series of experiments that showed the essential role of air in the growth of green plants. Priestley hypothesis: Plants restore to the air whatever breathing animals and burning candles.

9. Correct Option: (B) Jan Ingenhousz

Oxygen is evolved during photosynthesis as demonstrated by Jan Ingenhousz through an experiment on the aquatic plant -Hydrilla. He showed that in bright sunlight, small bubbles formed around the green parts of the plant while in dark, no bubbles were formed. He identified those bubbles as oxygen.

10. Correct Option: (C) Engelman

Engelmann used oxygen loving bacteria to grow on the algae. The bacteria congregating in the region of the red and blue light indicated that there was more oxygen being produced in these areas, indicating there was more photosynthetic activity in these areas.

11. Correct Option: (A) Pelletier & Caventou
Chlorophyll was first extracted by Pelletier and Caventou in 1817. At the time of their work, scientists know that green plants consume carbon dioxide and release oxygen.
12. Correct Option: (B) 40-100
Chloroplasts are present in large numbers in the cells (around 40 – 100) of plant tissue like the stomatal guard cells and ground tissues.
13. Correct Option: (A) Daniel I. Arnon
In the 1950s, Arnon performed research with Mary Belle Allen and F. Robert Whatley on chloroplasts and their role in photosynthesis, identifying a process which they named "photosynthetic phosphorylation."
14. Correct Option: (B) Photolysis
The splitting of water molecules is called photolysis of water. During the light reaction of photosynthesis, the absorption of light by the chlorophyll results in the loss of the electron from it. To fulfill the electron requirement, the water molecule splits and donates the electron to the chlorophyll and at that time oxygen is released.
15. Correct Option: (B) ATP, NADPH
The assimilatory powers produced during light reaction of photosynthesis are ATP and NADPH. These are utilized to form carbohydrates (sugars) during light independent reactions of photosynthesis.
16. Correct Option: (A) Less than even 1/1000 of a second
1/1000th seconds because there is A server-containing system.
17. Correct Option: (C) Blue & Red
Chlorophyll is a photosynthetic pigment. Violet-blue and Orange-red light wavelengths are absorbed by chlorophyll A. The rate of photosynthesis is more in blue and red light.
18. Correct Option: (B) $C_6H_{12}O_6$
The end product of this reaction is that the Co2 is converted into glucose (Carbohydrates form).
19. Correct Option: (A) Amylase
Salivary amylase (Ptyalin) is a glucose-polymer cleavage enzyme that is produced by the salivary glands.
20. Correct Option: (A) Bile juice
Emulsification is done by the action of bile secretes from the liver. Bile contains bile pigment and bile salts. Bile salts break down the fats into smaller globules and result in a milky emulsion. These emulsified fats are again digested by pancreatic lipase into fatty acid and glycerol.
21. Correct Option: (D) Both a & b
Pancreatic juice consists of alkaline (chiefly bicarbonate) fluid and enzymes; 200–800 ml is produced each day. Enzymes, such as trypsin, chymotrypsin, lipase, and amylase, are essential for the digestion of most of the protein, fat, and carbohydrate in the meal.
22. Correct Option: (C) Hydrochloric acid
Gastric juice is a unique combination of hydrochloric acid (HCl), lipase, and pepsin.
23. Correct Option: (D) All the above
We can avoid indigestion by having a simple well-balanced diet, eating food in a leisurely manner, and thoroughly masticating food and taking violent exercise after eating food.
24. Correct Option: (A) Kwashiorkor
Kwashiorkor is a type of malnutrition characterized by severe protein deficiency. It causes fluid retention and a swollen, distended abdomen. Kwashiorkor most commonly affects children, particularly in developing countries with high levels of poverty and food insecurity.

25. Correct Option: (A) Riboflavin (Vitamin B2)

Vitamin B2 deficiency (ariboflavinosis) can cause glossitis, along with angular cheilitis, cheilosis, peripheral neuropathy and other signs and symptoms. The glossitis in vitamin B2 deficiency is described as magenta.

26.Correct Option: (C) Vitamin D

This vitamin should be a part of your diet to help your fracture heal. It helps your blood take in and use calcium and build up the minerals in your bones.

27.Correct Option: (B) Vitamin K

Vitamin K is necessary for blood clotting, and it prevents excessive bleeding. This is because vitamin K helps to produce proteins required for blood coagulation. If a person has a deficiency of vitamin K it increases the time taken by the blood to clot. Severe deficiency of vitamin K causes death due to excessive blood loss in case of a cut or an injury.

28.Correct Option: (B) Folic Acid

Folic acid is an essential nutrient necessary for protein and nucleic acid synthesis (DNA and RNA). The production and regulation of these biological macromolecules are essential for survival and replication of organisms.

29.Correct Option: (C) Vitamin A, D, E, K

Vitamins A, D, E, and K are called fat-soluble vitamins, because they are soluble in organic solvents and are absorbed and transported in a manner similar to that of fats.

30.Correct Option: (C) Ingestion →Digestion → Absorption →Assimilation →Egestion

1) Ingestion - It includes the processes that take place while the food is in the mouth also known as a buccal cavity (such as chewing and grinding) and swallowing of the food - which sends it onward down the digestive tract.

2) Digestion - It is the process of breaking down food into simpler substances that can then be absorbed and assimilated into the tissues of the body.

3) Absorption - The digested food is absorbed into the bodily fluids from the alimentary canal. The process of absorption takes place in the small intestine with the help of villi (they are minute finger-like processes lined against the walls of the small intestine).

4) Assimilation - the process by which chemicals from food are taken into the cells of the body after digestion and absorption. The food absorbed by blood is transported to various parts of the body by the blood vessels.

5) Egestion - the process of removing undigested food material from the body through the anus.

Match the Following

1. Correct Matching is: a - R, b - Q, c - T, d - P, e – S

a) Light reaction – (R) Grana
b) Dark reaction – (Q) Stroma
c) Photolysis – (T) Release of Oxygen
d) Photon – (P) Smallest energy of light
e) Stomata – (S) Exchange of Gases

2. Correct Matching is: a - R, b - Q, c - P, d - S, e - T

a) C. B. Van Neil – (R) Formulated equation of photosynthesis
b) Jan Ingenhousz – (Q) Release of oxygen from Hydrilla plants
c) Engelmann – (P) Detected the point of maximum rate of photosynthesis
d) Pelletier & Caventou – (S) Observed chlorophyll in chloroplast
e) Daniel I. Arnon – (T) Extracted chloroplast

3.Correct Matching is: a - R, b - P, c - S, d - T, e – Q

a) Salivary glands – (R) Ptyalin
b) Stomach – (P) Gastric Juice
c) Liver – (S) Bile Juice
d) Pancreas – (T) Trypsin, Amylase, Lipase
e) Small Intestine – (Q) Intestinal juice

4.Correct Matching is: a - R, b - Q, c - S, d - P, e - T

a) Cyanocobalamin (B12) – (R) Pernicious anemia
b) Ascorbic Acid (C) – (Q) Scurvy
c) PhylloQuinone (K) – (S) Blood Clotting problems
d) Riboflavin (B2) – (P) glossitis
e) Niacin (B3) – (T) Pellagra

LEVEL-I

1. Correct option: (A) Ptyalin

It breaks starch into maltose and dextrin. It will break large, insoluble starch molecules into soluble starches (amylodextrin, erythrodextrin, and achrodextrin) to produce successively smaller starches and ultimately maltose.

2. Correct option: (C) Protein

Trypsin is an enzyme that helps us digest protein. In the small intestine, trypsin breaks down proteins, continuing the process of digestion that began in the stomach. It may also be referred to as a proteolytic enzyme, or proteinase.

3. Correct option: (D) Glucose, Glucose

Maltose, also known as malt, is a disaccharide made up of two alpha D-glucose units. The two units of glucose are linked with an alpha 1,4 glycosidic bond.

4. Correct option: (B) Lipase

Lipases are digestive enzymes that catalyse the hydrolysis of fat (lipid, triglyceride) into free fatty acids (FFA) and monoglycerides, molecules that can be absorbed into the bloodstream.

5. Correct option: (B) Insulin, Glucagon

The production of pancreatic hormones, including insulin, somatostatin, gastrin, and glucagon, play an important role in maintaining sugar and salt balance in our bodies.

6. Correct option: (A) Mouth

Saliva contains enzyme Ptyalin or Salivary amylase which helps digest starch present in food into dimeric maltose which is made up of glucose units.

7. Correct option: (B) Liver

The liver is the largest gland in the human body. It is also the largest (internal)organ in our body and can weigh up to 1.5 kg for a human adult. That is, about 1/50th of the body weight is because of the liver.

8. Correct option: (A) protein into peptones

Trypsin is an enzyme produced by the pancreas which breaks down protein molecules into simple and small protein molecules called proteoses and peptones.

9. Correct option: (D) Milk

Rennin is a proteolytic enzyme found in the gastric juice in the stomach of infants which helps in the digestion of Casein (milk protein). Rennin is also known as Chymosin synthesized by the chief cells.

10. Correct option: (A) Peristalsis

Peristalsis is a series of wave-like muscle contractions that move food through the digestive tract.

11. Correct option: (D) 2123/2123

The dental formula of adult human teeth is 2123/2123. It means that there are 2 incisors, 1 canine, 2 premolars and 3 molars in each half of the upper and lower jaw.

12. Correct option: (C) Epiglottis

Epiglottis is a flap-like structure present in the larynx which covers the respiratory tract when food enters the oesophagus. It is the function of epiglottis to prevent the entry of food into the respiratory tract.

13. Correct option: (C) Enamel

Enamel is the hardest substance of the human body, and it is the outer layer of the crown. Dentine is a layer that lies beneath the enamel of the tooth.

14. Correct option: (C) Mouth

The digestion of starch starts in the mouth and then continues in the intestine. The food in the mouth is chewed and mixed with salivary amylase to help in the easy movement of the food in the esophagus.

15. Correct option: (C) Production of Insulin

Insulin is a hormone, produced by beta cells of islets of Langerhans of pancreas.

16. Correct option: (C) Conversion of Glucose into Glycogen

Glycogenesis, the formation of glycogen, the primary carbohydrate stored in the liver and muscle cells of animals, from glucose. Glycogenesis takes place when blood glucose levels are sufficiently high to allow excess glucose to be stored in liver and muscle cells.

17. Correct option: (D) Sodium glycolate and taurocholate

In humans, taurocholic acid and glycocholic acid (derivatives of cholic acid) and Tauro chenodeoxycholic acid and glycochenodeoxycholic acid (derivatives of chenodeoxycholic acid) are the major bile salts.

LEVEL-II

1. Correct option: Correct option: (B) Stenson's duct

A duct, called Stensen's duct, drains saliva from the parotid gland into the mouth, at the area of the upper cheeks.

2. Correct option: (B) Peptic ulcers

In humans, excessive stimulation of the Vagus nerve disrupts the mechanism of hydrochloric acid secretion. Excessive secretion of gastric acid causes peptic ulcers. So, excessive stimulation of the Vagus nerve in humans may lead to peptic ulcers.

3. Correct option: (D) Bile Salts

Bile is a watery greenish fluid containing bile salts, bile pigments, cholestrol and phospholipid. Blue salts play an important role in digestion of fats. Therefore, in their absence emulsification of fat cannot take place.

4. Correct option: (D) Number and type of teeth in one half of both jaws

It showcases the peculiar pattern of arrangement, the kind of teeth present, and the number of a particular tooth in the mouth, at a given age. The dental formula constitutes the number of teeth in their respective position as they are present in the jaws. It's 2123/2123 per half of the jaw.

5. Correct option: (D) K

It is Vitamin K2, a component of vitamin K, synthesized by bacteria present in the intestine and it helps to treat osteoporosis.

6. Correct option: (C) Premolar and Molar

The premolar and molars are called cheek teeth. These are also called lophodont teeth. Their free ends are flattened and bear transverse ridges to crush and grind the food.

7. Correct option: (C) Night Blindness – Retinol – Carrot and milk

Night blindness (in which it is difficult or impossible to see in relatively low light) is one of the clinical signs of vitamin A deficiency and is common during pregnancy in developing countries. Retinol is the main circulating form of vitamin A in blood and plasma, and is present in carrots, green vegetables, and dairy products.

8. Correct option: (A) HCl

Ptyalin is a form of amylase found in the saliva of humans and some other animals. It converts starch into dextrin and maltose. HCl of the gastric juice inactivates or destroys salivary amylase or ptyalin in the stomach is well known to physiologists and physiological chemists.

9. Correct option: (C) Third molar, two in number

Humans have 12 molars, six in each jaw. Four of those are wisdom teeth, which is also called the third molar, which come in between the ages of 17 to 25.

10. Correct option: (A) Watery stool with blood and mucus

Patients with colitis present with watery diarrhea, abdominal pain, tenesmus, urgency, fever, tiredness, and blood in the stool.

11. Correct option: (C) 2 ATP

In anaerobic respiration, the ATP is only produced from Glycolysis as the NADH produced is used again in fermentative pathways. So only 2 ATPs are produced during Substrate level phosphorylation.

12. Correct option: (B) Ileum

Ileum is the part of the small intestine. Its mucosa is raised into numerous microscopic projections called the villi. The villi contain minute blind ended lymph vessel lacteals, which help in the absorption of fats.

13. Correct option: (B) Cholecystokinin

Cholecystokinin (in fact it is cholecystokinin pancreozymin) is a hormone which is secreted by the small intestine and stimulates pancreas to secrete and release digestive enzymes in pancreatic juice. Secretin causes the release of bicarbonates in the pancreatic juice.

14. Correct option: (A) Thiamine (B1)

Vitamin-B1 (thiamine) deficiency is most observed in alcoholic person. Prolonged deficiency of thiamine may lead to decreased mental function double vision reduced muscular contraction, lacking in decision capability. These all the ill effect results in a disorder known as Wernicke's syndrome.

15. Correct option: (C) Intestine

Duocrinin is a gastrointestinal hormone that is secreted within the intestine and stimulates the secretory activity of the duodenal glands (Brunner glands). Cholecystokinin is synthesized by the cells of the mucosal epithelium of the small intestine and secreted in the duodenum. It causes the release of digestive enzymes and bile from the pancreas and gallbladder, respectively.

16. Correct option: (B) Vitamin K

Hemorrhagic disease is a bleeding problem that occurs in a baby during the first few days of life. Babies are normally born with low levels of vitamin K, an essential factor in blood clotting. A deficiency in vitamin K is the main cause of hemorrhagic disease in newborn babies.

17. Correct option: (D) 38 ATP

In aerobic respiration, the full oxidation of one glucose molecule can theoretically yield 38 ATP molecules. But the net gain is always 36 ATP as 2 ATP has been previously used during glycolysis.

18. Correct option: (B) Milk changing casein into calcium para caseinate at pH 1-3

Rennin is found in gastric juice of infants which causes coagulation of milk and is the main component for the digestive process of infants. In the presence of calcium, this enzyme converts casein of milk into paracasein which is then treated by pepsin. Hence, rennin acts on milk proteins and converts casein into paracasein. Rennin has an optimum pH of 3 for proteolysis.

19. Correct option: (D) Maltose – Maltase

Maltase is an enzyme that hydrolyzes Maltose into glucose monomers in the intestine of humans. Rennin is a protein digesting enzyme found in infants that helps in digestion of milk protein casein. Starch is digested by amylase in the mouth. Whereas steapsins/ lipases are responsible for fat breakdown. Chymotrypsin is also a proteases that digests proteins in peptides and peptones.

20. Correct option: (C) A is true but R is false

Numerous tube-like glands are present in the wall of the small intestine which secrete intestinal juice into the intestinal lumen. This juice contains several enzymes like enterokinase, amino peptidases, peptidases, maltase, lactase, lipase etc. for digesting various types of food. On the other hand, the enzymes RNase and DNase are present in the pancreatic juice.

21. Correct option: (D) A is false but R is true

The large intestine is responsible for the absorption of water and minerals. The mucosa has many goblet cells but does not have any villi.

22. Correct option: (A) Both A and R are true, and R is correct explanation of A

The peristaltic movement of the gut is aided by the activity of muscles. The food material is mixed with the enzymes in the gut when the muscle contracts and relaxes. Different glands in the alimentary canal release these enzymes. This helps proper food digestion in the small intestine.

23. Correct option: (A) Both A and B are true, and R is the correct explanation of A

Carbohydrates are more suitable to produce energy in the body than proteins and fats because carbohydrate molecules contain relatively more oxygen than the others, and consequently, require less molecules of oxygen for their oxidation. Carbohydrates are also stored in the tissue as glycogen for use in the production of energy, when necessary.

24. Correct option: (C) A is true, but R is false

In the small intestine Pancreatic amylase converts starch and dextrose into maltose and small intestine is main site for digestion of carbohydrates.

25. Correct option: (C) A is true, but R is false

About 30% of starch is hydrolyzed in the oral cavity by the action of salivary amylase into a disaccharide- maltose. Sucrase hydrolyses sucrose into glucose and fructose.

Breathing and Exchange of Gas

LEVEL-I

1. Correct option: (D) b and c

The plain dwellers when move to the higher altitudes certain physiological changes occur in order to cope up with decreased oxygen levels. The important changes are increased breathing rate, increased red blood cells productions with a decreased oxygen binding capacity.

2. Correct option: (1A) bicarbonate
Approximately seventy percent of carbon dioxide absorbed by the blood will be transported to the lungs as bicarbonate ions.
3. Correct option: (A) 1000-1200 mL
Total oxygen that can be carried by blood in 1000-1200 ml.
4. Correct option: (A) Both assertions and reason are true, and the reason is the correct explanation of the assertion
5. Correct option: (D) 1%-2%
Approximately 98% of total oxygen transported in the blood is bound to haemoglobin, while only 2% is dissolved directly in plasma.
6. Correct option: (A) HCO_3 bicarbonate ions
About 70% of CO2 is converted into bicarbonate ions and are transported in plasma.
7. Correct option: (C) Nostrils →Pharynx →Larynx →Trachea →Bronchi→Bronchioles →Alveoli
When you inhale through your nose or mouth, air travels down your pharynx (back of your throat), passes through your larynx (voice box) and into your trachea (windpipe). Your trachea is divided into two air passages called bronchial tubes. One bronchial tube leads to your left lung, the other to your right lung.
8. Correct option: (C) Diaphragm and external intercostal muscles contract
During inhalation, the diaphragm contracts and moves downwards. This increases the space in your chest cavity into which the lungs expand. The intercostal muscles between the ribs also help in enlarging the chest cavity. Whereas during exhalation the diaphragm relaxes and moves upwards into the chest cavity.
9. Correct option: (A) CO
Carbon monoxide combines with hemoglobin to form carboxyhemoglobin at any or all of the oxygen-binding sites of hemoglobin, and also acts to increase the stability of the bond between hemoglobin and oxygen, reducing the ability of the hemoglobin molecule to release oxygen bound to other oxygen-binding sites.
10. Correct option: (A) Sodium bicarbonate
Carbon dioxide can be transported through the blood via three methods. It is dissolved directly in the blood, bound to plasma proteins or hemoglobin, or converted into bicarbonate. The majority of carbon dioxide is transported as part of the bicarbonate system.
11. Correct option: (D) 4.5%
Inhaled air has about 0.04% of carbon dioxide. This is also the amount of carbon dioxide present in the atmosphere. The exhaled air contains all the carbon dioxide that is a by-product of the cellular metabolic processes. Hence, the percentage of carbon dioxide in the exhaled air is 4.5%.

12. Correct option: (B) Alveoli
Both the lungs in humans are filled with tiny balloon-like structures called air sacs or alveoli. Their membrane is highly thin and is surrounded by many capillaries. Alveoli are the actual sites where gaseous exchange takes place between the air in the lungs and blood.
13. Correct option: (C) Epiglottis
The epiglottis is an elastic cartilaginous leaf-shaped flap covering the opening of the larynx.
14. Correct option: (B) Thoracic cavity

Your lungs are located in your chest (your thorax). Your thoracic cavity is the name of the space that contains your lungs and other organs. Your lungs rest on a muscle called your diaphragm.

15. Correct option: (A) As a solution, carbamino compounds, bicarbonates

There are three means by which carbon dioxide is transported in the bloodstream from peripheral tissues and back to the lungs: (1) dissolved gas, (2) bicarbonate, and (3) carbaminohemoglobin bound to hemoglobin (and other proteins).

16. Correct option: (C) Tidal Volume

Tidal volume is the volume of air normally inspired or expired in one breath without any effort. It is about 500 ml in the average human adult male.

17. Correct option: (C) Envelops the lungs

A pleura is a serous membrane that folds back on itself to form a two-layered membranous pleural sac. The outer layer is called the parietal pleura and attaches to the chest wall. The inner layer is called the visceral pleura and covers the lungs, blood vessels, nerves, and bronchi.

18. Correct option: (B) More surface area for diffusion of gases.

There are approximately 600 – 800 million alveoli in the two lungs of a human being. The alveoli are sac-like structures that store the air that one breathes in. The alveoli are stretchable.

19. Correct option: (A)a – r; b – s; c – q; d – p

Column – I		**Column – II**
a) Larynx	-	r) voice box
b) Trachea	-	s) Windpipe
c) Alveoli	-	q) air sacs
d) Epiglottis	-	p) Lid of larynx

LEVEL-II

1. Correct option: (A) Sigmoid

The oxygen-hemoglobin dissociation curve is a graphical representation of the relationship between the amount of oxygen bound to hemoglobin and the partial pressure of oxygen in the blood. The curve is sigmoidal, with a steep slope at low partial pressures of oxygen and a more gradual slope at higher partial pressures.

2. Correct option: (D) Hyaline cartilage

Trachea is lined with incomplete rings of hyaline cartilage.

3. Correct option: (C) Reduces oxygen carrying capacity of blood

CO is not added to tobacco but is formed when tobacco is burned incompletely. This happens when there is too little oxygen present to convert all of the carbon in the tobacco into harmless carbon dioxide. Cigarette smoke can contain large quantities of CO. CO has higher affinity than O2 for hemoglobin.

4. Correct option: (B) Cartilaginous rings

The trachea is supported by c-shaped rings of cartilage which hold them in place when there is less air inside it. Due to this, the walls of trachea do not collapse even when there is a little air. So, the correct option is 'cartilaginous rings.

5. Correct option: (B) Jerky incomplete inspiration

A hiccup can be described as a jerky incomplete inspiration. It is an involuntary contraction of the diaphragm that may repeat several times per minute. In medical terms it is known as synchronous diaphragmatic flutter SDF

6. Correct option: (C) CO_2 concentration in artery blood

Thus, the respiratory centres that are present in the brain are stimulated by the carbon dioxide content in the arterial blood.

7. Correct option: (A) Bicarbonate; 70%

During transport, most of CO 2 is passed into erythrocytes. A major part (nearly 70%) forms carbonic acid on reacting with water in erythrocytes and then form bicarbonate.

8. Correct option: (B) Several hundred grams

In the blood of an adult man the total content of haemoglobin is roughly several hundred grams

9. Correct option: (D) 150 mL

Anatomical dead space is the volume of the conducting airways (from the nose, mouth, and trachea to the terminal bronchioles). These conduct gas to the alveoli, but no gas exchange occurs here.

10. Correct option: (A) 16%

Inhaled air contains approx. 21%oxygen, 0.04% carbon dioxide and 79%nitrogen while exhaled air contains approx. 16% oxygen, 4% carbon dioxide and 79% nitrogen.

11. Correct option: (B) Kept floating

A child was killed through asphyxiation. Postmortem confirmed it because a piece of lung put in water kept floating.

12. Correct option: (B) Cigarette smoking

Emphysema is lung condition that causes shortness of breath due to damage of air sacs in the lungs (alveoli). The main cause of emphysema is a long–term exposure to air–born irritants including tobacco smoke, air pollution, dust.

13. Correct option: (C) Residual Volume.

Residual volume is the amount of air that remains in the lungs after forceful expiration. It is about 1,500 ml.

14. Correct option: (C) Directly proportional to the concentration of CO in the medium

The binding of one CO molecule to haemoglobin increases the affinity of the other binding spots for oxygen, leading to a left shift in the dissociation curve. This shift prevents oxygen unloading in peripheral tissue and therefore the oxygen concentration of the tissue is much lower than normal.

15. Correct option: (C) blood buffer

The bicarbonate buffering system of blood is an important buffer system in the acid-base homeostasis of living things, including humans. As a buffer, it tends to maintain a relatively constant plasma pH and counteract any force that would alter. In this system, carbon dioxide CO2 combines with water H2O to form carbonic acid H2CO3, which in turn rapidly dissociates to form hydrogen ions H+ and bicarbonate HCO3 as shown in the reactions below.

$CO_2 + H_2O \rightleftarrows H_2CO_3 \rightleftarrows HCO_3^- + H^+$

The carbon dioxide - carbonic acid equilibrium is catalysed by the enzyme carbonic anhydrase; the carbonic acid - bicarbonate equilibrium is simple proton dissociation/association and needs no catalyst. Any disturbance of the system will be compensated by a shift in the chemical equilibrium.

16. Correct option: (A) As a solution, carbamino compounds, bicarbonates

There are three means by which carbon dioxide is transported in the bloodstream from peripheral tissues and back to the lungs: (1) dissolved gas, (2) bicarbonate, and (3) carbaminohemoglobin bound to hemoglobin (and other proteins).

17. Correct option: (A) To regulate inhalation

The pneumotaxic center, located in the upper pons, sends inhibitory impulses to the inspiratory center, terminating inspiration, and thereby regulating inspiratory volume and respiratory rate. This center likely is involved in the fine-tuning of breathing.

18. Correct option: (D) Chloride shift

Chloride shift (also known as the Hamburger shift or Hamburger phenomenon, named after Hartog Jakob Hamburger) is a process which occurs in a cardiovascular system and refers to the exchange of bicarbonate ($HCO3^-$) and chloride (Cl^-) across the membrane of red blood cells (RBCs).

19. Correct option: (A) Relaxes and arches.

When you breathe out, or exhale, your diaphragm and rib muscles relax, reducing the space in the chest cavity. As the chest cavity gets smaller, your lungs deflate, similar to how air releases from a balloon.

20. Correct option: (C) Diaphragm and external intercostal muscles contract

During inspiration, the dome shaped diaphragm stretches and become flat and increases the size of the thoracic chamber. As a result, the pressure in the thorax is reduced to less than atmospheric pressure. Air, therefore, enters the lungs.

21. Correct option: (A) Chloride shift

The chloride content of RBCs increases when oxygenated blood becomes deoxygenated. This is temed as chloride shift or hamburger shift.

22. Correct option: (D) Common cold by corona virus

SARS Coronavirus 2, an RNA virus causes severe pulmonary distress conditions infecting the mucosal tissues of lungs, resulting in difficulty in breathing and life-threatening conditions. Coronaviruses accounts for approx. 20% of common cold viral infections.

23. Correct option: (C) nostrils-pharynx-larynx-trachea-bronchi-bronchioles-alveoli

When you inhale through your nose or mouth, air travels down your pharynx (back of your throat), passes through your larynx (voice box) and into your trachea (windpipe). Your trachea is divided into two air passages called bronchial tubes. One bronchial tube leads to your left lung, the other to your right lung.

24. Correct option: (D) a-i, b-ii, c-iii, d-iv

Silicosis is an interstitial lung disease caused by breathing in tiny bits of silica, a common mineral found in many types of rock and soil. Over time, exposure to silica particles causes permanent lung scarring, called pulmonary fibrosis.

Emphysema develops over time and involves the gradual damage of lung tissue, specifically the destruction of the alveoli.

During an asthma attack, the smooth muscle layer goes into spasm, narrowing the airway causing difficulty in breathing.

Bronchitis is a condition that develops when the airways in the lungs, called bronchial tubes, become inflamed and cause coughing, often with mucus production.

25. Correct option: (B) a – ii; b – iii; c – iv; d – i

Vital capacity (VC) is the maximum amount of air a person can expel from the lungs after a maximum inhalation. It is equal to the sum of inspiratory reserve volume, tidal volume, and expiratory reserve volume. The total lung capacity (TLC), about 6,000 mL, is the maximum amount of air that can fill the lungs (TLC = TV + IRV + ERV + RV). The maximum volume of air that can be inspired after reaching the end of a normal, quiet expiration. It is the sum of the tidal volume and the inspiratory reserve volume (TV + IRV). Expiratory Capacity (EC) Total volume of air a person can expire after a normal inspiration. It includes tidal volume and expiratory reserve volume (TV+ERV).

Transportation

LEVEL-I

1. Correct option: (A) Fish

A closed-type circulatory system is seen in fish. Blood circulates via a series of vessels in a closed circulatory system without coming into direct touch with human tissues. An open circulatory system, in contrast, allows blood to circulate more freely and directly bathe bodily tissues.

The circulatory systems of snails, cockroaches, and scorpions are all open. As a result, their blood does not benefit from blood arteries and travels freely through their bodily cavities. Although this circulatory system is not as effective as a closed system, it is enough for these tiny creatures.

2. Correct option: (B) Lymphocytes and monocytes

White blood cells known as agranulocytes are distinguished by the lack of granules in their cytoplasm. About 20–30% of all white blood cells are this smallest subset of white blood cells. Both lymphocytes and monocytes are agranulocytes.

The generation of antibodies and immunological memory are two important functions of lymphocytes in the immune response. By phagocytosing pathogens and presenting antigens to other immune cells, monocytes, on the other hand, contribute to the immune system's defense against infections.

Granulocytes, such as eosinophils and basophils, have granules in their cytoplasm. Inflammation and allergic responses are both mediated by granulocytes.

3. Correct option: (B) Myogenic

In vertebrates, the heartbeat is myogenic, which implies that it develops inside the heart muscle itself. Pacemaker cells are specialized cells within the heart that produce electrical impulses that cause the heart muscles to contract, so starting the heartbeat. The sinoatrial node, or SA node, is found in the right atrium of the heart and contains these pacemaker cells. About 70 times every minute, the SA node produces an electrical impulse that causes the heart to beat. As a result of this contraction, blood is pumped into the ventricles, which contract and send blood to the rest of the body. Although it does not start the pulse, the nervous system can affect it by accelerating or decelerating it.

4. Correct option: (C) Right atrium and right ventricle

In the right atrium and right ventricle of the heart, there is a valve called the tricuspid. It stops the right ventricle's right atrium's blood from returning there. Three leaflets, or flaps of tissue, make up the tricuspid valve, which opens and closes to let blood flow in one direction.

The left atrium and left ventricle are where you'll find the bicuspid valve. Furthermore, it stops blood from returning from the left ventricle to the left atrium. Between the left ventricle and the aorta is where the aortic valve is situated. It stops the left ventricle's leftward return of blood from the aorta.

5. Correct option: (D) Ca

Coagulation, the process by which blood clots, depends heavily on calcium ions. They are necessary for a number of clotting cascade processes, such as the activation of clotting factors and the production of fibrin, which aids in forming a stable blood clot to stop excessive bleeding.

The remaining ions, sodium (Na), potassium (K), and magnesium (Mg), are not directly involved in blood coagulation. However, they can influence the activity of other ions or

proteins involved in the clotting cascade, which can have an indirect effect on blood clotting.

6. Correct option: (A) One minute

Cardiac output measures how much blood the heart pumps in a minute. It is computed by multiplying the heart rate (the number of beats per minute) by the stroke volume (the volume of blood pumped by each heartbeat). In order to quantify cardiac output, liters per minute are used. A healthy adult's resting heart rate is around 5 liters per minute. This indicates that the heart circulates the body with around 5 liters of blood each minute.

The other choices are incorrect. The cardiac index refers to the volume of blood that the heart pumps in a second. The term "stroke volume" refers to the volume of blood that the heart pumps during each beat. The minute ventilation refers to the volume of blood the heart pumps out per 100 heartbeats.

7. Correct option: (D) Pulmonary vein

In the pulmonary veins, oxygenated blood is located. These veins return blood that has been filled with oxygen from the lungs to the left atrium of the heart. To provide the body's tissues and organs access to oxygen, this oxygenated blood is subsequently pushed into the systemic circulation.

From the right ventricle of the heart to the lungs, deoxygenated blood travels by the pulmonary artery. Deoxygenated blood from the body enters the right atrium and right ventricle.

8. Correct option: (C) 0.8 seconds

The cardiac cycle in man takes about 0.8 seconds. The heart needs this amount of time to go through one full cycle of contraction and relaxation. Systole and diastole are the two stages of the heart cycle.

The stage of contraction is called systole. The heart muscle contracts and pumps blood out of the heart during systole. The relaxation phase is known as diastole. The heart muscle relaxes and fills with blood during diastole.

The pacemaker cells in the heart control the cardiac cycle. These cells produce electrical impulses that contract and relax the heart muscle. The sinoatrial node (SA node), which is found in the right atrium of the heart, is where the pacemaker cells are situated.

9. Correct option: (B) bilirubin

Blood coagulation is not something that the word "bilirubin" may be used for. Heme from haemoglobin is broken down to produce bilirubin. It is then broken down by the liver and eliminated as bile, giving faces its distinctive brown color. Blood coagulation and clotting are not directly affected by bilirubin.

The blood coagulation process also involves the other words, fibrin, fibrinogen, and calcium. A clot is created when the protein fibrin creates a web that enfolds platelets and blood cells. The starting point for fibrin is fibrinogen. The activation of numerous proteins involved in the coagulation cascade requires calcium.

10. Correct option: (D) All the above

The thymus, tonsils, and lymph nodes are only a few human organs that contain lymphoid tissue. These organs are crucial parts of the immune system that aid in the development of immune cells as well as the filtration and surveillance of the lymphatic fluid for infections and foreign objects.

T lymphocytes (T cells), which are essential for adaptive immunity, are matured in the thymus. By capturing and recognizing possible infections coming via the mouth and

throat, tonsils act as defense systems. Filtering lymphatic fluid is crucial because lymph nodes contain immune cells that hunt out and destroy dangerous chemicals. The body's defense against infections and disorders is ensured by the lymphoid tissues as a whole, which support immunological responses.

11. Correct option: (A) Both 'A' and 'R' are true, and 'R' is correct explanation of 'A'.

The absence of clotting proteins in blood serum accounts for the fact that it does not have the ability to clot. When blood clots, clotting proteins like fibrinogen cause the blood to solidify and create a mass. Serum is the liquid fraction of blood that is left over after the clotting process has eliminated the clotting factors; as a result, serum lacks the ability to clot. Therefore, the reason (R) correctly explains why the assertion (A) is true.

12. Correct option: (B) Calcium ions and platelet factor

Blood coagulation depends on calcium ions and platelet factors. The activation of clotting factors and the production of fibrin are two phases in the clotting cascade activated by calcium ions. When platelets come into touch with injured tissue, platelet factors are released from the platelets. They cause the coagulation cascade's other proteins to bind with one another, causing a blood clot to develop.

Anticoagulants, such as heparin, oxalates, and citrates, stop blood from clotting. They are thus not necessary for the coagulation of blood.

13. Correct option: (C) Blood group is designated based on the presence of antibodies in the blood plasma.

Instead, then considering the presence of antibodies in the blood plasma, blood groups are really determined by the presence or absence of certain antigens on the surface of red blood cells. Option 3 is wrong because it depicts the methodology used to determine blood groups incorrectly. The presence or absence of antigens like A, B, and Rh(D) on the surface of red blood cells determines a person's blood type, and the compatibility of organ transplants and blood transfusions depends on these antigen-antibody interactions.

The other claims are accurate. Antigens on the surface of red blood cells, not antibodies in blood plasma, are used to determine blood group. A person with type A blood can get blood from a type B donor. A person with type A blood cannot, however, donate blood to someone with type B blood. A person with blood type AB may accept blood from any blood type.

14. Correct option: (B) 1 – c, 2 – a, 3 – d, 4 –b

1)Lymphatic system - c. To drain back the tissue fluid to the circulatory system

The lymphatic system helps to maintain fluid balance by collecting and returning extra tissue fluid (lymph) to the circulatory system as well as moving immune cells.

2) Pulmonary vein - a. Carries oxygenated blood

The pulmonary vein facilitates the delivery of oxygen from the lungs to the rest of the body by transporting oxygenated blood from the lungs to the left atrium of the heart.

3) Thrombocytes - d. Coagulation of blood

Platelets called thrombocytes are necessary for the coagulation of blood. When there is an injury, they are crucial in the formation of blood clots that stop excessive bleeding.

4) Lymphocytes - b. Immune Response

Platelets called thrombocytes are necessary for the coagulation of blood. When there is an injury, they are crucial in the formation of blood clots that stop excessive bleeding.

15. Correct option: (A) Action mentioned in Statement 1 is dependent on action mentioned in Statement 2

According to Statement 1, the atria get blood from every area of the body, which then flows to the ventricles. As noted in Statement 2, the coordinated action potential produced at the Sino-Atrial (SA) node aids in the flow of blood from the atria to the ventricles. The electrical impulse that causes atrial contraction and assists in blood pumping into the ventricles is started by the SA node. As a result, Statement 1 and Statement 2 both depend on the electrical signal produced at the SA node in order for blood to flow from the atria to the ventricles.

LEVEL-II

1. Correct option: (A) Left atrium → left ventricle → aorta

Left atrium: The oxygenated blood from the lungs enters the left atrium.
Left ventricle: The left atrium pumps the blood into the left ventricle.
Aorta: The left ventricle pumps the blood into the aorta, the main artery of the body.
Arteries: The aorta branches into smaller arteries, which carry the blood to all parts of the body, including the hand.
The other answer choices are incorrect because they do not show the correct path of the blood flow.

2. Correct option: (B) Only (ii) is false

Statement (i) is true: About 6-8 percent of the blood plasma is made up of proteins. Albumin, which serves to regulate blood pressure and transport nutrients and hormones throughout the body, is the protein that makes up the majority of plasma.
Statement (ii) is false: Mineral content is not very high in plasma. In actuality, it only has approximately 0.9% mineral content. In plasma, sodium, chloride, potassium, and bicarbonate are the minerals that are most prevalent.
Statement (iii) is true: Serum is plasma minus the clotting components. Proteins called clotting factors aid in blood clotting. To avoid the blood clotting in the test tube when blood is collected for testing, the clotting factors are eliminated. Serum is the resultant liquid.
Statement (iv) is true: Since these substances are constantly moving throughout the body, they are also present in plasma. These materials are delivered by the plasma to the cells, where they are used for cellular development, energy production, and repair.

3. Correct option: (B) Only ii

An open circulatory system has an unregulated blood flow to many organs. This is due to the fact that there is no method to regulate the blood's flow to certain organs, and blood flows freely through the bodily cavities.
Due to the blood not being restricted to arteries but rather flowing freely through the bodily cavities, the blood pressure in an open circulatory system is significantly lower than in a closed circulatory system. Small blood channels called capillaries link veins and arteries. They are absent from open circulatory systems. In an open circulatory system, the blood returns slowly to the heart. This is due to the heart's inability to pump blood through the body.

4. Correct option: (B) Both assertion and reason are true, but reason is not the correct explanation of assertion

The claim and the justification are both true. White blood cells (WBCs) do travel from the circulation via the endothelium (inner lining) of blood vessels to the site of damage or infection, a process known as diapedesis, to collect at the site of wounds. The justification given, nevertheless, is not the one that adequately explains the claim.

White blood cells actively squeeze through endothelial cell gaps during diapedesis, however the phrase "squeezing from the endothelium" may be deceptive. The primary idea is that white blood cells may go to the site of inflammation or infection thanks to diapedesis, which enables them to exit the circulation.

5. Correct option: (D) A – iii, B – iv, C – i, D – ii

A. Plasma - iii. Fluid part of blood

The liquid portion of blood is called plasma. Water, electrolytes, hormones, waste materials, and numerous proteins are all present in it.

B. Red blood cells - iv. Carry oxygen

Erythrocytes, another name for red blood cells, are in charge of returning carbon dioxide to the lungs for expiration and transporting oxygen from the lungs to the body's tissues.

C. White blood cells - i. Eat up the bacteria

Leukocytes, commonly known as white blood cells, are important components of the immune system. They support the body's defence mechanisms against diseases and outside aggressors including germs and viruses.

D. Platelets - ii. Coagulation of blood

The process of blood coagulation or clotting involves platelets. By generating clots at the site of an injury, they aid in controlling bleeding and limiting excessive blood loss.

6. Correct option: (C) Only iv

i. This statement is incorrect. Instead of ventricles, the human heart's upper two chambers are known as atria. The atria are where the heart's blood is received.

ii. This statement is correct. Ventricles are the actual names of the lowest two chambers of the human heart. Blood is pumped from the heart through ventricles.

iii. This statement is correct. The septum, not the pericardium, serves as the barrier between the heart's chambers. A two-layered sac called the pericardium protects the heart by surrounding it.

iv. This statement is incorrect. Deoxygenated blood is transported via the pulmonary artery from the right ventricle to the lungs, where it is oxygenated. After receiving oxygen in the lungs, the blood travels back to the heart through the pulmonary veins.

7. Correct option: (D) A, B, AB and O

Blood group AB is the consequence of a blood group 'A' (genotype AO) person marrying a blood group 'B' (genotype BO) person. This happens because the possible kid can receive the A allele from the first parent and the B allele from the second parent. In addition, the child may receive the O allele from the second parent and the A allele from the first parent, resulting in blood group A. They might also acquire the blood type B allele from their second parent and the blood group O allele from their first parent. Last but not least, if both parents inherit the O gene, the child would have the blood group O.

8. Correct option: (A) Both A and B antigens on RBC but no antibodies in the plasma

Red blood cells (RBCs) from people with blood type AB are coated with both A and B antigens. This implies that they won't experience an immunological reaction to the transfusion if they get blood from donors of any blood group.

They don't have any plasma antibodies to A or B antigens. They won't be able to develop an immune response to entering blood that contains A or B antigens due to this absence of antibodies.

The A or B antigens are not present on the red blood cells of people with blood type O. Additionally, they do not possess either anti-A or anti-B antibodies. However, they are only able to accept blood from O blood types. This is due to the fact that if they receive

blood from someone with blood group A or B, their body may create antibodies to the A or B antigens, which may result in a response known as a hemolytic transfusion reaction.

9. Correct option: (D) A – Pulmonary vein-takes impure blood from body parts, PO2 = 60 mm Hg

The correct label and its functions among the given options are Pulmonary vein - takes impure blood from body parts, PO2 = 60 mm Hg

Oxygenated blood is returned to the heart by pulmonary veins from the lungs.

After being oxygenated in the lungs, the blood has a comparatively greater oxygen concentration (PO2 = 60 mm Hg) in the pulmonary veins.

The pulmonary veins contain oxygen-rich blood that has been oxygenated in the lungs, making it relatively pure in terms of oxygen concentration. As a result, the phrase "impure blood" does not apply to them.

10. Correct option: (D) Type O

Blood type O, sometimes referred to as the universal donor blood type, can be safely used to transfuse a person with unknown blood group who has lost a significant amount of blood. People of any ABO blood type can have blood type O because it lacks A or B antigens on the surface of the red blood cells.

Blood from a type O donor can be safely transfused in an emergency when the recipient's blood type is unknown.

11. Correct option: (B) Only i

Blood pressure is the force that the heart's pumping action places on the blood arteries as it travels through them. Systolic blood pressure refers to the force exerted by blood when blood vessels constrict, while diastolic blood pressure refers to the force exerted when blood vessels relax. 120/80 is the typical blood pressure reading.

Statement i is true. This number for blood pressure is extremely high and may harm kidney and brain blood vessels. It can cause significant health issues like renal disease and a stroke if left untreated.

Statement ii is false. The 130/90 mm Hg blood pressure reading is not very high and may be controlled by meditating and drinking some cold liquids. There's no need for therapy for this.

Statement iii is false. Adults should have a blood pressure of 120/80 mm Hg. A measurement of 100/55 mm Hg is seen as being on the low side and might be an indication of low blood pressure.

Statement iv is false. Low blood pressure is defined as having a measurement of 105/50 mm Hg or below. You could experience fatigue, dizzy, and lightheadedness from low blood pressure. Additionally, it may make it challenging to focus and think effectively. It would not necessarily make someone "very active."

12. Correct option: (A) i and iii

Statement i is true because arteries have thicker walls and a smaller diameter than veins. This occurs so that the arteries, which transport oxygenated blood from the heart to the rest of the body, can handle the pressure of the blood. Conversely, veins do not require as much strength since they just transport deoxygenated blood from the rest of the body back to the heart.

Statement iii is accurate since calcium ions are crucial for blood coagulation. A number of clotting factors, which are proteins that aid in stopping bleeding, must first be activated by calcium ions.

Statement ii is incorrect Because only individuals of blood group AB may donate blood to them. This is due to the fact that individuals with blood type AB contain both A and

B antigens on their red blood cells. When they receive blood from someone with a different blood type, their immune system will fight the foreign antigens and result in a transfusion response.

13. Correct option: (D) Only D

Adult human red blood cells (RBCs) have a special property called enucleation that gives them extra room for transporting oxygen. RBCs can increase their haemoglobin content, which is in charge of binding and transporting oxygen throughout the body, by eliminating the nucleus and other organelles. RBCs are better able to carry out their main role because to this adaptation, which also improves oxygen transport efficiency. The bone marrow produces RBCs, which are then discharged into the circulation and have a lifespan of roughly 120 days. They are then disassembled and recycled. RBCs are somatic cells, which means they are cells that are part of the tissues that make up the body. Because they lack the organelles and enzymes required for metabolism, RBCs cannot metabolize.

14. Correct option: (B) P – Artery, Q – Vein, R - Capillary

P – Artery - Arteries are blood arteries with strong walls that transport oxygenated blood from the heart. They feature a thick layer of smooth muscle and a limited lumen (the area through which blood flows). This enables them to resist the intense blood pressure caused by the heart's pumping action.

Q – Vein - Veins are blood channels with thin walls that return anaemic blood to the heart. Compared to arteries, they have a broader lumen and a thinner coating of smooth muscle. As a result, the blood may return to the heart more easily.

R - Capillary - The body's tiniest blood vessels are called capillaries. Because of their extremely thin walls, the cells can receive oxygen and nutrients from the blood. Arterioles and veins are linked by capillaries

15. Correct option: (C) Both the statements are correct

Statement 1: When blood clots, serum, liquid rich in proteins, separates off. Red blood cells, white blood cells, platelets, and plasma all make up blood. The platelets and fibrinogen in the plasma create a network known as fibrin during blood clotting. Red and white blood cells become trapped in this net, creating a clot. The liquid that remains after the clot develops is called serum. Proteins including albumin, globulins, and fibrinogen are abundant in it.

Statement 2: An insoluble protein called fibrin is created during blood clotting from fibrinogen. The plasma contains fibrinogen, a soluble protein. When blood clots, fibrin, a mesh made of platelets and plasma fibrinogen, is formed. The white blood cells and red blood cells get caught in this net, creating a clot. Because the fibrin is insoluble, it does not disintegrate in water. This is what gives the clot its hard, solid texture.

16. Correct option: (D) S

Blood that has lost its oxygen content travels from the heart to the lungs through the pulmonary artery. As the blood has not yet been oxygenated in the lungs, the oxygen level in the blood in the pulmonary artery is quite low. Of all the blood arteries in the body, the blood in the pulmonary artery will contain the least amount of oxygen.

The chart shows that sample S has the lowest oxygen content, so it is most likely taken from the pulmonary artery.

17. Correct option: (B) Larger veins

Through the bigger veins, lymph is cleansed and its beneficial components are reintroduced into the circulatory system. Larger bodily veins eventually get the contents of lymphatic channels. This is how the lymphatic system helps to keep the fluid balance in the body while moving trash, bacteria, and other materials out of tissues.

Blood is transported from the heart through arteries. They receive no lymph. The tiniest blood vessels are capillaries. They serve as the point where the blood and tissues exchange nutrients and oxygen. They receive no lymph. A muscular organ, the heart circulates blood throughout the body. It receives no lymph.

18. Correct option: (C) i , iii and iv

Statement i is true. Platelets are sometimes referred to as thrombocytes.
Statement ii is false. Because they are anucleated, platelets lack a nucleus. Additionally, their blood volume is substantially larger than that of 6000–8000 mm3, with an average of 150,000–400,000 platelets per mm3.
Statement iii is true. They assist in the blood's coagulation. For blood to clot, platelets are required. When a blood artery is damaged, platelets congregate to create a plug that aids in stopping the bleeding.
Statement iv is true. A decrease in their population may result in clotting issues, which may cause the body to lose large amounts of blood. Additionally true is this. Thrombocytopenia is a condition that can result from a decrease in platelet production. Thrombocytopenia may result in severe bleeding, which is potentially fatal.

19. Correct option: (D) i, ii, iii and iv

A set of processes known as the cardiac cycle help the blood to flow through the heart. Systole, which denotes the contraction of the cardiac muscles, and diastole, which denotes their relaxation, are the two primary phases of it (ii). Specialized nodal tissues, namely the sinoatrial node (SA node) and atrioventricular node (AV node), which control the heart's rhythm and coordination, orchestrate the cardiac cycle (iii). Changes in the length of the cardiac cycle have a direct effect on the cardiac output, causing an increase or reduction in the volume of blood the heart pumps per unit of time (iv). As a result, each of the assertions (i, ii, iii, and iv) offered accurately describes a different element of the cardiac cycle.

20. Correct option: (B) A vein starts from an organ and ends up in another organ.

An intricate circulatory system known as a "portal system" involves the passage of blood through two capillary networks before it returns to the heart. In this system, a vein emerges from one organ, where it draws blood from capillaries, before transporting it to a different organ's second set of capillaries and ultimately returning to the heart. Before the blood is returned to the heart, this configuration enables an effective interchange of nutrients, gases, or other things between two distinct organs. The definitions of the other choices (A, C, and D) do not adequately convey the idea of a gateway system.

Excretory System

LEVEL-I

1. Correct option: (B) Nephron

In the human excretory system, the nephron is the excretory unit. It is a tiny organ located in the kidneys and is in charge of filtering blood, eliminating waste, and controlling the fluid balance in the body. Millions of nephrons are found inside each kidney, and they are essential for preserving the body's equilibrium by removing waste materials, extra ions, and water from the blood to make urine. The urine that the nephron produces then travels through the urinary system before being expelled from the body. The body's interior environment is maintained and kept in balance thanks to this effective filtration and waste disposal procedure.

The fundamental element of the nervous system is the neuron. Signals are sent by them throughout the body. Invertebrates excrete their waste through nephridia. Worms, insects, and snails are just a few of the many species that have them. The excretory system of flatworms is the flame cell. They are specific cells that filter blood and expel trash.

2. Correct option: (B) Kidney → ureters → bladder → urethra

After the nephrons in the kidneys filter and remove waste products, excess water, and ions from the blood, the resulting fluid, called urine, is then transported through the ureters, which are muscular tubes connecting each kidney to the bladder. The urine is temporarily stored in the bladder until it reaches a certain level of fullness. When the bladder is filled, the urine is expelled from the body through the urethra, a tube that leads from the bladder to the external opening. This process allows the body to efficiently eliminate waste and maintain the proper balance of fluids and ions.

3. Correct option: (B) Vasopressin

Vasopressin, also known as antidiuretic hormone (ADH), has a direct impact on urination. Produced in the hypothalamus and released by the posterior pituitary gland, vasopressin regulates the water balance in the body by controlling the reabsorption of water in the kidneys. When vasopressin levels are high, the kidneys reabsorb more water from the urine, resulting in concentrated urine and decreased urine output. Conversely, when vasopressin levels are low, more water is excreted in the urine, leading to increased urine output and potentially more diluted urine. This hormone plays a crucial role in maintaining proper fluid balance and preventing excessive dehydration or water retention.

Urination is not directly influenced by estrogen, FSH, or adrenal hormones. Cortisol and adrenaline are only two of the chemicals that the adrenal gland generates. Although this is an indirect consequence, cortisol can cause an increase in urine output. Urine output can also be increased by adrenaline; however, this normally only happens in stressful circumstances. FSH is a hormone that affects how many eggs women produce. Women's ovaries generate the hormone known as estrogen. By influencing the bladder, FSH and estrogen can both indirectly affect urination. They don't, however, directly affect urine.

4. Correct option: (A) Glomerular filtration, Tubular reabsorption, Tubular secretion

Glomerular filtration: In order to create urine, this is the initial step. A network of capillaries in the nephron called the glomerulus is where it takes place. Known as the Bowman's capsule, the glomerulus is encircled by this structure. Blood pressure drives water and other tiny solutes into the Bowman's capsule as blood passes through the glomerulus. Water, glucose, amino acids, salts, and waste materials can be found in this filtrate.

Tubular reabsorption: The second stage in the production of urine is this. The renal tubule, a protracted, winding tube that is a component of the nephron, is where it develops. Cells having unique characteristics that enable them to selectively reabsorb particular compounds from the filtrate line the renal tubule. Reabsorbed substances include salts, water, glucose, and amino acids. The waste is not reabsorbed.

Tubular secretion: Urine formation proceeds via this stage. A renal tubule hosts it. During tubular secretion, certain chemicals are actively transferred from the blood into the filtrate. Potassium ions, certain medicines, and hydrogen ions are some of these compounds.

5. Correct option: (C) Bowman's capsule

Glomerulus and Bowman's capsule make comprise the renal corpuscle. The Bowman's capsule is a component of the nephron that is located in the cortex of the outer zone of the kidney. A system of capillaries called the glomerulus filters blood. It acts as the first site of filtration when blood components are filtered into the renal tubules and encloses the glomerulus, a network of capillaries. By removing waste and water from the blood, this mechanism starts the creation of urine.

The medulla, the kidney's inner region, contains the proximal convoluted tubule (PCT), the distal convoluted tubule (DCT), and the loop of Henle, whereas the cortex, the kidney's outer region, has the bowman's capsule.

6. Correct option: (A) Cortex

The kidney's cortical region is its outermost region. It produces urine and filters blood, and it is dark brown in color. The proximal and distal convoluted tubules, which are involved in the reabsorption and secretion processes, as well as the glomeruli, which filter blood, are among the many structures found in the renal cortex. The nephrons, the kidney's microscopic filtering units, are housed in this area of the kidney, which is crucial in the early stages of urine generation.

The renal pyramids are what make up the medulla. The renal pyramids are radially organized cone-shaped structures. Urine is gathered from the renal tubules via collecting ducts found in the renal pyramids. The calyces, which are cup-shaped structures that collect urine from the renal pyramids, are where the pee travels after leaving the collecting ducts. The ureter, a tube that transports urine from the kidney to the bladder, is where the calyces then empty into.

7. Correct option: (A) Proximal convoluted tubule

In the renal system, most glucose molecules are reabsorbed in the proximal convoluted tubule (PCT). This tubule is situated in the cortex of the kidney and is responsible for the initial reabsorption of various valuable substances, including glucose, amino acids, ions, and water, from the filtrate. Glucose is actively transported from the PCT back into the bloodstream, ensuring that the body retains this essential energy source rather than excreting it in the urine. This process helps maintain proper blood glucose levels and efficiently utilizes nutrients during urine formation.

Reabsorbing water and concentrating the urine are two functions of the distal convoluted tubule, loop of Henle, and collecting ducts. However, for reabsorbing glucose, the PCT is the most crucial component of the nephron.

8. Correct option: (B) Waste products

When the kidneys are unable to eliminate waste materials from the blood, uremia develops. This may occur as a result of acute renal injury, chronic kidney disease, or other kidney-damaging disorders.

Waste materials accumulate in the blood when the kidneys are unable to filter them out. Numerous symptoms, such as weariness, nausea, vomiting, lack of appetite, muscular cramps, itching, and mental disorientation, may result from this. Uremia can be lethal in extreme circumstances.

Urea, creatinine, uric acid, and potassium are among the waste products that accumulate in the blood in uremia. The liver creates urea as a waste product as a result of protein breakdown. A waste product produced by the muscles is creatinine. The breakdown of purines results in the waste product uric acid. Potassium is a crucial mineral that supports the control of the heartbeat and other physiological processes. Therefore, uremia is a condition in which the body retains waste materials rather than excess water.

9. Correct option: (B) Vasopressin

Vasopressin, sometimes referred to as antidiuretic hormone (ADH), boosts reabsorption in the nephron's collecting ducts. Vasopressin, which is produced in the hypothalamus and released by the posterior pituitary gland, works on the collecting ducts to increase their water permeability. This concentrates the urine and helps the body conserve water by allowing the body to reabsorb more water from the urine back into the bloodstream. This system is necessary to keep the body's fluid balance in check, avoid dehydration, and modify urine concentration in response to hydration requirements.

Renin, aldosterone, and insulin don't make the collecting ducts reabsorb more. Renin stimulates angiotensin II production, which raises blood pressure. Aldosterone increases sodium reabsorption and potassium excretion, which regulates blood pressure and electrolyte balance. By encouraging the absorption of glucose into cells, insulin aids in the regulation of blood sugar levels.

10. Correct option: (D) Increased water consumption

Urine output rises as a direct result of increased water intake. The human body's fluid equilibrium is upset when you drink more water, therefore the extra water needs to be expelled. An increase in urine production is the consequence of the kidneys being prompted to filter and eliminate the extra water and waste.

Increased amounts of exercise can cause perspiration, which can cool the body and lessen the need to urinate. Sweating and reduced urine output are two additional effects of elevated body temperature. Evidently, less water consumption will result in less urine production.

11. Correct option: (B) Podocytes

The Bowman's capsule is an essential part of the nephron of the kidney, which is in charge of the early filtering of blood to create urine. A single layer of specialized cells known as podocytes lines the inside surface of the Bowman's capsule. Larger molecules, including as proteins, are prevented from entering the first filtrate by the intricate foot-like "pedicels" of these podocytes, which wrap around the glomerular capillaries.

The other options are incorrect. The alternate choices are wrong. Some plants' cells contain crystals in the form of needles called raphides. Red blood cells are known as erythrocytes. Dopocytes are not a class of cell.

12. Correct option: (C) Ultra filtration and selective reabsorption

Ultrafiltration, in which blood is filtered through the glomerulus in Bowman's capsule to generate a filtrate, is a step in the nephron's process of producing urine. Small compounds, ions, and water is all present in this filtrate. Then, selective reabsorption occurs, where waste materials and excess substances are left in the developing urine while necessary chemicals like glucose, ions, and water are reabsorbed from the filtrate back into the circulation. The mechanisms involved in producing urine in the nephron are not correctly described by the other alternatives (A, B, and D).

13. Correct option: (C) Distal convoluted tubule

The distal convoluted tubule (DCT) is in charge of preserving the ideal pH and concentration of urine. It is the section of the renal tubule that links to the collecting ducts following the Henle loop. Further ion reabsorption and secretion of water, other chemicals, and other processes take place in the DCT, allowing for fine-tuning of urine composition.

The majority of the glucose, amino acids, and other crucial nutrients from the filtrate are reabsorbed by the proximal convoluted tubule (PCT). The Henle loop is a U-shaped

structure that aids in urine concentration. All of the kidney's nephrons' urine must be collected and sent to the bladder through the collecting duct.

14. Correct option: (A) Vasopressin

Vasopressin, commonly known as antidiuretic hormone (ADH), is the primary factor that causes urine to concentrate in the collecting ducts. When the blood osmolarity varies, the hypothalamus produces vasopressin, which is then released by the posterior pituitary gland. As a result of its action on the collecting ducts, which increases their water permeability, more water may be reabsorbed from the developing urine and returned to the circulation. Concentrated urine is the outcome.

The hormone insulin is in charge of controlling blood sugar levels. Blood pressure and salt levels are controlled by aldosterone. The response of "fight or flight" is brought on by adrenaline.

15. Correct option: (A) 300 – 700 ml

The urinary bladder may hold 300 to 700 ml of fluid on average. The bladder is a muscular hollow organ that may expand to hold various volumes of pee. The neurological system receives a signal from the bladder's stretch receptors when urine fills it, which causes the urge to urinate.

The other options (B, C, and D) do not accurately represent the common range of bladder storage capacity.

16. Correct option: (A) 1.5 to 1.7 liters

Each day, the nephrons produce around 180 liters of glomerular filtrate in total. To keep osmotic equilibrium and preserve vital nutrients, only a small fraction of this filtrate—roughly 188 to 188.5 liters per day—is reabsorbed back into the blood vessels. Urination serves as the body's method of removing the residual nephron residue, which contains nitrogenous waste products. An individual produces around 1.5 liters of urine on average each day as a result. This range of urine production falls within 1.5 to 1.7 liters per day.

17. Correct option: B) Charles Hufnage

In 1954, an American surgeon named Dr. Joseph Murray carried out the first successful kidney transplant on a person. On December 23, 1954, the procedure was carried out at the Peter Bent Brigham Hospital in Boston, Massachusetts. Ronald Herrick, a 23-year-old man with end-stage renal failure, was the receiver. Richard Herrick, his identical twin brother, served as the donor.

William Harvey was an English doctor who made the discovery of blood circulation. American surgeon Charles A. Hufnagel created the first prosthetic heart valve in the early 1950s. American agricultural scientist Norman Borlaug received the Nobel Peace Prize in 1970 for his efforts to boost grain production.

18. Correct option: (A) Bilirubin, biliverdin and urochrome

The breakdown products of hemoglobin from dead red blood cells include bilirubin and biliverdin. The substance bilirubin, which gives bruises their yellow color, is finally broken down in the liver and eliminated in bile. The pigment urochrome, which is produced when heme is broken down, is part of what gives urine its yellow color. Despite not being formed from hemoglobin, uric acid and urea are metabolic wastes as well. These wastes are likewise carried to the kidneys, where they are eliminated through the urine. Although they are not metabolic wastes, water and mineral salts are nonetheless expelled in the urine.

19. Correct option: (C) Adrenal gland

Each kidney's upper surface is occupied by the endocrine glands known as the adrenal glands The adrenal glands are made up of two parts: the cortex and the medulla. The

cortex produces a number of hormones, including cortisol, aldosterone, and androgens. The medulla produces the hormone adrenaline. The body's metabolism, stress response, and electrolyte balance are just a few of the physiological processes that these hormones play a significant part in controlling.
In the front of the neck, the thyroid gland is located. An organ behind the stomach is called the pancreas. Both endocrine and exocrine actions are performed by it. The pituitary gland, which regulates several other endocrine glands, is sometimes referred to as the "master gland." It can be found at the bottom of the brain.

20. Correct option: (B) Vasopressin

The illness known as diabetes insipidus is brought on by a lack of the hormone vasopressin, often referred to as antidiuretic hormone (ADH). The hypothalamus makes ADH, which the pituitary gland then releases. By increasing the water permeability of the collecting ducts, it aids in the kidneys' ability to retain water. Vasopressin regulates the reabsorption of water in the kidneys, which is a key factor in managing the water balance. A lack of vasopressin causes the kidneys to be unable to effectively concentrate urine, which causes excessive thirst and urination.
A hormone called insulin is produced by the pancreas. It facilitates the body's utilization of glucose as a fuel. The adrenal glands create a hormone called adrenaline. In times of stress or peril, it is released. The parathyroid glands create the hormone known as parathormone. It aids in controlling blood calcium levels.

21. Correct option: (B) Ureotelism

The removal of urea, a byproduct of metabolism, is called ureotelism. In animals, including humans, this is the most typical way that nitrogenous waste is excreted. Ammonia, a poisonous gas created when proteins break down, is converted to urea in the liver. The urea is subsequently carried to the kidneys, where it is eliminated as urine.
The removal of ammonia-based metabolic waste is known as ammoniotelism. Several aquatic creatures, including fish, excrete nitrogenous waste in this basic form. The removal of uric acid-based metabolic waste is known as uricotelism. This is how birds, reptiles, and insects expel nitrogenous waste. Aminotelism is not a true method of eliminating nitrogenous waste.

22. Correct option: (C) Constant composition of the blood irrespective of the nature of the food or fluid intake

While the juxtaglomerular apparatus releases renin to maintain glomerular blood flow and pressure, the kidney's antidiuretic hormone (ADH) controls blood and fluid volume. The renin-angiotensin-aldosterone pathway, on the other hand, is blocked by the atrial natriuretic factor, which lowers blood pressure through vasodilation. In order to maintain constant blood composition regardless of fluid consumption, the kidneys' osmoregulation system uses ADH to regulate water balance. Together, these activities help to keep the body's internal environment consistent in terms of composition, fluid volume, and blood pressure.

23. Correct option: (A) Nephrons

Nephrons are the smallest functional unit of the kidney. They are responsible for filtering the blood and producing urine. Each kidney contains about 1 million nephrons. The nephron is the kidney's smallest functioning unit. Filtering blood, reabsorbing necessary nutrients, and excreting waste to generate urine are all tasks carried out by nephrons. The glomerulus, Bowman's capsule, proximal convoluted tubule, loop of Henle, distal convoluted tubule, and collecting ducts are just a few of

the components that make up each nephron. These components all work together to carry out the many kidney activities.
The other options (B, C, and D) are components of the nephron or larger structures associated with kidney function.

24. Correct option: (D) Both A and B

The rate at which blood plasma is filtered via the glomeruli (small filtering units) of the kidneys is known as the glomerular filtration rate (GFR). It's an essential indicator of kidney health since it shows how well the kidneys are removing waste and extra chemicals from the blood to create urine.
The correct GFR reading is around 125 mL/min per kidney, or 180 liters per day when the two kidneys are combined.
So, in this case, if the options are presented as "Both (A) and (B)," the correct answer is D) Both (A) and (B) because both the values 125 mL/min and 180 L/day are accurate representations of the glomerular filtration rate.

25. Correct option: (B) Aldosterone

An enzyme in the adrenal cortex called aldosterone is responsible for its production. When blood potassium levels are high or the sodium level is low, it is discharged. In order to enhance salt and water absorption and reduce potassium excretion, aldosterone operates on the kidneys. The fluid and electrolyte balance of the body is preserved as a result.
ADH, also known as vasopressin, is a hormone that controls salt reabsorption.
Enzymes called renin and rennin are part of the renin-angiotensin-aldosterone system (RAAS). The reabsorption of sodium is not directly impacted, though.

26. Correct option: (D) ADH

The posterior pituitary gland houses the hormone ADH, also known as antidiuretic hormone, which is generated by the brain. When the concentration of dissolved chemicals in the blood is high or the blood volume is low, it is discharged. Water is conserved and dehydration is prevented because to ADH's effect on the kidneys, which increases water absorption.
Other options are incorrect. They are not involved in reabsorption of water. Growth hormone, also known as STH, is produced by the pituitary gland. The pituitary gland creates ACTH, also known as adrenocorticotropic hormone. It causes the adrenal cortex to generate cortisol when stimulated. The pituitary gland produces luteinizing hormone (LH).

27. Correct option: (C) Both (A) and (B)

Na+ and Cl- are both removed by the ascending limb of Henle's loop and the proximal convoluted tubule. Na+ and Cl- can pass through the ascending limb of Henle's loop, but water cannot. In contrast to water, Na+ and Cl- are actively carried out of the tubule. As a result, a concentrated salt solution forms in the kidney's medulla, which aids in luring water from the collecting duct and into the circulation.
Additionally permeable to Na+ and Cl-, the proximal convoluted tubule also possesses active transport systems for reabsorbing glucose, amino acids, and other nutrients. Na+ and Cl- are still excreted from the tubule, although not in the same quantities as in the ascending limb of Henle's loop.
The descending limb of Henle's loop is impermeable to Na+ and Cl-, but it is permeable to water. This means that water is drawn out of the tubule into the medulla of the kidney, which helps to concentrate the salt solution.

28. Correct option: (B) Kidney

Columns of Bertini are extensions of the cortex of the kidney that project into the medulla. They are made up of connective tissue and blood vessels. They help to support the medulla and to increase the surface area of the kidney. They help to support the medulla and to increase the surface area of the kidney.

Column of Bertini is not found in liver, ovaries, and testes.

29. Correct option: (D) Na^+ ions

Aldosterone is a hormone that is produced by the adrenal glands. It helps to regulate the balance of sodium ions in the body. It acts on the distal convoluted tubule and collecting ducts of the nephron to increase the reabsorption of sodium ions, which in turn affects water reabsorption and blood pressure regulation.

Keto acids, glucose, and K+ ions are not reabsorbed by aldosterone.

30. Correct option: (A) Cortex

Malpighian corpuscles, also known as renal corpuscles, are located in the renal cortex of the kidney. They consist of Bowman's capsule and the enclosed glomerulus, and they play a crucial role in the initial filtration of blood to form the glomerular filtrate.

The other options are incorrect. The kidney's medulla is situated inside the cortex. The testes are where the germ cells are found.

31. Correct option: (C) Liver

The liver generates urea as a waste product. Amino acids that are no longer required by the organism are broken down by the liver. Ammonia and carbon dioxide are the by-products of this breakdown. The liver turns ammonia into urea because it is harmful. Urine is produced when urea is delivered to the kidneys and discharged there.

The urinary bladder, alimentary canal, and kidneys are all involved in the process of urine formation, but they do not synthesize urea.

LEVEL – II

1. Correct option: (D) Nearly 99 percent of the glomerular filtrate is reabsorbed by the renal tubules

Filtering the blood and getting rid of waste is done by the kidneys. The capillary network that filters blood is known as the glomerulus. The majority of the water and nutrients are then reabsorbed in the renal tubules after the filtrate has been passed through them. Only 1% of the filtrate is eliminated in the urine.

Although it is really impermeable to water, the ascending limb of the Loop of Henle actively transfers electrolytes (sodium, potassium, and chloride) out of the tubule. Water can pass through the descending limb of the Loop of Henle and be reabsorbed by osmosis, which concentrates the filtrate. Bicarbonate ions (HCO3) can be reabsorbed by the distal convoluted tubule to assist control the body's acid-base balance.

2. Correct option: (A) Hydrogen ions are actively secreted into the filtrate

Human urine is usually acidic due to the active secretion of hydrogen ions (protons) into the filtrate within the renal tubules. This production of hydrogen ions aids in maintaining the body's acid-base homeostasis and pH balance. By selectively reabsorbing or secreting hydrogen ions and bicarbonate ions, the kidneys play a critical part in preserving the body's acid-base balance, helping to control blood pH. The urine's overall acidity is a result of the active release of hydrogen ions into the filtrate.

The other options are incorrect. The peritubular capillaries' sodium transporter does not switch hydrogen ions for sodium ions. Instead, it swaps sodium ions for chloride ions. This aids in preserving the blood's osmolarity. Plasma proteins that are excreted are not acidic. In actuality, they have a mild alkaline pH. Acidity is not produced by the

interchange of potassium and sodium. In actuality, it guards against alkalosis and buffers the blood.

3. Correct option: (D) i, ii, iii and iv

i. The outer layer of the kidney is called capsule. This is true. The renal capsule is the name of the kidney's outermost covering. It is a fibrous coating that shields the kidney from harm.

ii. Cortex is divided into outer cortex and inner medulla. This is true. The inner medulla and the outer cortex make up the kidney's structure. The proximal and distal convoluted tubules, together with Bowman's capsules and glomeruli, are located inside the cortex.

iii. Medulla is divided into medullary pyramids. This is true. The kidney's interior layer is called the medulla. The pyramids of the medulla are separated from it. Conical-shaped structures called the medullary pyramids extend into the renal pelvis.

iv. The cortex extends in between the medullary pyramids which is called as columns of Bertini. This is also correct. In between the medullary pyramids, the cortex expands. The columns of Bertini are a protrusion of the cortex. The medullary pyramids are supported in part by the Bertini columns.

4. Correct option: (D) A – iv, B – iii, C – ii, D – i

A. PCT - iv. Reabsorbing about 90% of the important buffer HCO_3 from filtrate

The initial component of the nephron is called the proximal convoluted tubule (PCT). About 60% of the water and 90% of the glucose, amino acids, and other tiny molecules from the filtrate are reabsorbed by it. The PCT is also involved in the hemostasis of Na+ and K+. It does not, however, absorb HCO3.

B. Descending loop of Henle - iii. Permeable to water but not to salt.

The second portion of the nephron is called the ascending Henle loop. Water cannot pass through it, but NaCl can. This enables the NaCl to accumulate in the kidney's medulla, aiding in the formation of a concentration gradient that stimulates water reabsorption in the ascending loop of Henle.

C. Ascending loop of Henle - ii. Permeable to NaCl but impermeable to water.

The third component of the nephron is the ascending Henle loop. Salt cannot pass through it, but water can. As a result, more water may be reabsorbed from the filtrate, increasing the concentration of the filtrate. It does not, however, work in the hemostasis of Na+ and K+.

D. DCT - i. Functions in Na+ & K+ hemostasis.

Fourth in the nephron structure is the distal convoluted tubule (DCT). It is in charge of secreting ions like K+ and H+ as well as reabsorbing roughly 10% of the water. Additionally, the DCT reabsorbs 90% of the crucial buffer HCO3 from filtrate.

5. Correct option: (C)

The concentration of insulin in the bloodstream is uniformly distributed throughout the body, indicating a consistent distribution throughout the whole bloodstream. Insulin continually decreases from the blood when it is metabolized and used, demonstrating a constant rate of elimination. When the blood concentration is graphed over time, the drop is exponential. In contrast, insulin is eliminated from the circulation by urinary excretion, resulting in an initial decrease in urine concentration that progressively rises over time as insulin is filtered and eliminated. As a result, it is anticipated that the graph of urine insulin concentration vs time would follow the opposite pattern of the blood concentration graph, beginning low and gradually rising.

6. Correct option: (D) i and iv

Loop of Henle (i): A vital component of the nephron that greatly aids in the production of concentrated urine is the Loop of Henle. It produces an osmotic gradient in the kidney medulla, allowing for the active transport of ions (such as sodium and chloride) in the ascending leg and the reabsorption of water from the filtrate in the descending limb. The concentration of the urine depends on this gradient.

Vasa Recta (iv): In the kidney's medulla, parallel to the Loop of Henle, are specialized capillaries known as the vasa recta. The osmotic gradient created by the Loop of Henle is maintained in large part by them. The interchange of water and solutes is possible in the vasa recta without interfering with the concentration gradient, which helps keep the osmotic gradient from dissipating.

Even while Bowman's capsule (iii) and the glomerulus (ii) are both engaged in the early filtering of blood to create the filtrate, neither is directly in charge of the concentration of urine. The vasa recta maintain the concentration process, which is principally driven by the reabsorption and control of water and solutes in the Loop of Henle.

7. Correct option: (B) Only 3

Urine is hypertonic in distal convoluted tubule. This is incorrect. The urine is actually hypotonic in the distal convoluted tubule.

When the urine passes into collecting tubule, it becomes hypertonic. This is also incorrect. The urine becomes more hypotonic as it passes into the collecting tubule.

Urine is isotonic in proximal convoluted tubule. This is correct. The urine is isotonic in the proximal convoluted tubule.

Urine becomes more and more hypotonic as it passes through Henle's loop. This is also correct. The urine becomes more and more hypotonic as it passes through the ascending limb of the loop of Henle.

8. Correct option: (C) A is true and R is false

The assertion is true. Podocytes are specialized cells that line the interior of Bowman's capsule. To create a filtration barrier, the foot processes of podocytes interdigitate with one another. Water and tiny molecules can travel across this barrier without being blocked, but proteins and other big molecules cannot.

The reason is false. The surface area for absorption is not increased by the podocytes' projections. They expand the filtering surface area. The basement membrane and the podocyte foot processes combine to create the filtration barrier. The foot processes are secured to the basement membrane and tightly sealed together thanks to the projections of the podocytes. Larger molecules cannot cross the filtration barrier because of this seal.

9. Correct option: (C) A – Medullary pyramid, B – Calyx, C – Renal artery, D – Renal vein, E - Ureter

A) Medullary pyramid: The kidney's medulla contains triangle-shaped structures called medullary pyramids. They have channels for collecting urine, which are used to move it from the nephrons to the renal pelvis.

B) Calyx: Resembling a cup, the calyx absorbs urine from the medullary pyramids. The renal pelvis is reached by the union of many small calyces into large calyces.

C) Renal artery: From the heart to the kidneys, the renal artery transports oxygenated blood. Through the renal arteries, blood enters the kidneys, giving them with nutrients and oxygen.

D) Renal vein: Deoxygenated blood from the kidneys is transported via the renal vein to the inferior vena cava, which then returns the blood to the heart.

E) Ureter: The ureter is a tube that transports urine from the kidney's renal pelvis to the bladder, where it is temporarily stored until being eliminated.

10. Correct option: (B) Inside the kidney the cortical region extends in between the medullary pyramids as renal pelvis

Inside the kidney the cortical region extends in between the medullary pyramids as renal pelvis is incorrect. The renal pelvis, which is part of the collecting system that collects urine from the medullary pyramids and directs it into the ureter for delivery to the bladder, is situated in the middle of the kidney. The medullary pyramids are deeper structures in the medullary zone, whereas the cortical area is found on the kidney's outermost section. The renal pelvis is located deeper inside the kidney and is not an extension of the cortical area.

The other given statements are correct.

11. Correct option: (C) Ammoniotelic - Aquatic Amphibia; Ureotelic - Frog, Humans; Uricotelic - Pigeon, Lizards, Cockroach

Ammonia is the main nitrogenous waste that ammonotelic animals emit. Ammonia is extremely harmful, and it takes a lot of water to successfully dilute and expel it. Water is available to aquatic creatures, such as amphibians, allowing for effective ammonia excretion.

Animals that are ureotelic expel urea as their main nitrogenous waste. In addition to requiring less water for excretion, urea is less hazardous than ammonia. Both frogs and people are ureotelic. Urea is passed through the skin by frogs and the urine of humans.

Uricotelic animals excrete uric acid as their primary nitrogenous waste. Uric acid is relatively non-toxic and can be excreted with minimal water loss. Birds, reptiles (including lizards), and insects like cockroaches are uricotelic. Pigeons are also uricotelic birds.

12. Correct option: (B) Podocytes: Create minute spaces (slit pores) for the filtration of blood into the Bowman's capsule

Podocytes are specialized cells found in the nephron's Bowman's capsule. Pedicels, which are special foot-like extensions, are attached to them and wrap around the glomerulus capillaries. Filtration slits are the little apertures seen on these pedicels. The filtration barrier that enables tiny molecules and ions from the blood to be filtered into the Bowman's capsule is made up of these filtration slits, glomerular endothelial cells, and the basement membrane. The kidneys' first stage in producing urine is the first filtration of blood.

The other options are incorrect. The artery that supplies the glomerulus with blood is called the afferent arteriole. Blood is removed from the glomerulus through the efferent arteriole, which is an artery. The Henle's loop, a U-shaped component of the nephron, aids in the concentration of urine. A portion of the water and solutes from the filtrate are reabsorb able by the distal convoluted tubule of the nephron.

13. Correct option: (B) When someone drinks lot of water, ADH release is suppressed

The pituitary gland stores antidiuretic hormone (ADH), also known as vasopressin, which is a hormone generated by the brain. In reaction to elevated blood osmolality, ADH is released. The concentration of solutes in a solution is measured by osmolality. The hypothalamus receives a signal to release ADH when blood osmolality rises. ADH then goes to the kidneys where it increases the water permeability of the collecting ducts. As a result, the kidneys are able to reabsorb more water from the urine, increasing its concentration.

Option A is inaccurate since the blood's osmolality rises when the body loses a lot of water through evaporation. The kidneys reabsorb more water and create more

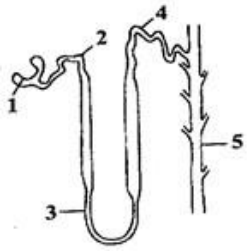
concentrated urine as a result of this increased ADH release. Because exposure to low temperatures does not induce the production of ADH, Option C is inaccurate. Because an increase in glomerular blood flow does not trigger the production of angiotensin II, Option D is untrue.

14. Correct option: (A) Uric acid can be excreted in almost solid form

The fact that uric acid may be expelled in a virtually solid state and is comparatively insoluble in water makes it advantageous to expel nitrogenous wastes in this form. This helps creatures with limited access to water because it enables them to preserve water by excreting waste that is concentrated. The main nitrogenous waste product emitted by animals like birds, reptiles, insects, and certain other species that have trouble conserving water is uric acid.

Option B is inaccurate since little energy is needed to create uric acid. Option C is inaccurate because uric acid is not the first byproduct of the metabolic breakdown of acids. Because uric acid is not eliminated through the lungs, Option D is untrue. It is eliminated by way of the kidneys and bladder.

15. Correct option: (C) 2 and 3



Both the descending loop of Henle and the proximal convoluted tubule (PCT) include cells with many microvilli and mitochondria. The PCT is the nephron's initial structure. The majority of the water, glucose, and amino acids in the filtrate are reabsorbed by it. The many microvilli seen in PCT cells improve the surface area for absorption. They also have a large number of mitochondria, which provide the process of absorption with energy.

The section of the nephron that descends into the kidney's medulla is known as the descending loop of Henle. It is in charge of making the pee more concentrated. Additionally lined with microvilli, the cells of the descending loop of Henle assist to improve the surface area for water absorption.

There are fewer microvilli and mitochondria in the Bowman's capsule, the distal convoluted tubule (DCT), and the collecting duct. This is due to the fact that they are not as effective in absorbing water and solutes.

16. Correct option: (D) a, b and c

a) The expulsion of urine stored in the urinary bladder is called micturition

This statement is true. The process of releasing urine from the bladder is known as micturition. It entails a synchronized set of actions, such as the internal urethral sphincter relaxing and the bladder muscles contracting to force urine out the body.

b) Urine is collected in the nephrons and flows into the ureters

The kidneys generate urine. After that, it is gathered in the renal pelvis and passes via the ureters. The muscular tubes known as ureters transport urine from the kidneys to the bladder.

c) Micturition can also be initiated voluntarily by contracting the abdominal muscle which applies pressure over the urinary bladder, activating the stretch receptors

The abdominal muscles can be consciously contracted to start urinating. The bladder experiences pressure as a result of the contraction of the abdominal muscles. The bladder wall's stretch receptors are activated by this pressure, which then sets off the micturition reflex.

17. Correct option: (C) Only i

The juxtaglomerular apparatus (JGA) is activated when the glomerular filtration rate (GFR) drops. The enzyme renin is released as a result of this activation. Angiotensin I is created through a sequence of events involving angiotensinogen that are triggered by

renin. The angiotensin-converting enzyme (ACE) subsequently converts angiotensin I into angiotensin II. Aldosterone is secreted as a result of the powerful vasoconstrictor angiotensin II stimulating the adrenal cortex. In the kidneys, aldosterone encourages the reabsorption of water and sodium ions.

Antidiuretic hormone (ADH) promotes water reabsorption from the tubule's last sections, avoiding diuresis. Angiotensinogen in the blood is converted by renin to angiotensin I, and then to angiotensin II. Due to its potent vasoconstrictor properties, angiotensin II raises GFR and glomerular blood pressure. Vasodilation (dilation of blood vessels) brought on by ANF can lower blood pressure.

18. Correct option: (C) Both statements are correct

Statement 1 is correct. Osmoregulation is the process of maintaining the quantity of water and dissolved solutes in balance in the body. This is important for a number of reasons, including maintaining blood pressure, cell volume, and electrolyte balance.

Statement 2 is correct. The excretion of water and dissolved solutes is achieved through organs, such as the skin and the kidneys. The skin helps to regulate water loss through sweating, and the kidneys help to regulate water and electrolyte balance by filtering blood and producing urine.

The major organs involved in osmoregulation are the skin and kidneys, however they don't function independently. Water and electrolyte balance is also maintained by other organs such the liver, lungs, and intestines.

19. Correct option: (B) Only 1 and 3

The capillary network of the glomerulus' capillary network is divided into several branches, while the rubber tubing at B is not. As the glomerulus is a cluster of capillaries that filters blood into the nephron, this is a significant defect in the concept. To reflect this, the rubber tubing should be divided into several branches.

To symbolize the tangled tubules of the nephron, the stem of the thistle funnel (E) is not coiled. The nephron's convoluted tubules are in charge of reabsorbing water and other solutes from the filtrate. To depict this, curl the stem of the thistle funnel.

The other statements in the question are not accurate. The rubber tubing at B should be continuous, but it should contain small pores to allow the filtration of blood into the nephron.

20. Correct option: (D) PCT, DCT and Malpighian corpuscle

In the cortical part of the kidney are the Proximal Convoluted Tubule (PCT), Distal Convoluted Tubule (DCT), and Malpighian corpuscle (which includes the glomerulus and Bowman's capsule).

In the cortical area of the kidney, there is a Malpighian corpuscle. It consists of a Bowman's capsule, which encloses the glomerulus, and a glomerulus, which is a ball of capillaries.

The PCT, which is likewise found in the kidney's cortical area, receives the filtrate thereafter. The PCT reabsorbs the majority of the electrolytes, amino acids, and glucose in the filtrate.

The DCT, which is likewise situated in the kidney's cortical area, receives the filtrate after that. Some water and electrolytes are reabsorbed by the DCT from the filtrate.

21. Correct Matching is: a-T; b-R; c-U; d-Q; e-S; f-P

Answer:

a) Glomerulus - T) Ultrafiltration
b) Loop of Henle - R) Concentrated urine
c) Proximal convoluted Tubule - U) Reabsorption of glucose, amino acids
d) Distal convoluted tubule - Q) Regulate K, Na and pH+

e) Collecting duct - S) Absorption of water
f) Vasopressin - P) Urine formation
a) Glomerulus - T) Ultrafiltration
The glomerulus, a network of tiny blood arteries (capillaries) in the kidney, is crucial to the ultrafiltration process. The process of making urine begins with ultrafiltration, in which blood is filtered under pressure to separate bigger molecules—such as proteins and blood cells—from smaller ones, such as waste particles, ions, and water. The glomerulus functions as a filter that lets smaller molecules through while keeping bigger ones in the circulation.
b) Loop of Henle - R) Concentrated urine
The kidney tubules' Loop of Henle is a part that establishes a gradient of concentration in the tissue around it. Urine must be concentrated using this concentration gradient. The Loop of Henle aids in the reabsorption of water and ions when urine passes through the tubules, which helps to concentrate urine by eliminating extra water from the filtrate.
c)Proximal convoluted Tubule - U) Reabsorption of glucose, amino acids
An essential component of the nephron is the Proximal Convoluted Tubule (PCT), which is where crucial chemicals like glucose and amino acids are reabsorb able. To avoid losing important nutrients in the urine, these compounds are selectively reabsorbed back into the circulation from the tubular fluid.
d) Distal convoluted tubule - Q) Regulate K, Na and pH+
The Distal Convoluted Tubule (DCT) is in charge of controlling the body's ion balance, including the ratio of potassium (K+) to sodium (Na+). The DCT controls the acid-base balance and adjusts the amounts of these ions, both of which are essential for proper physiological operation.
e) Collecting duct - S) Absorption of water
The Collecting Duct, the nephron's last component, is essential for reabsorbing water. The amount of water reabsorbed back into the circulation by the collecting duct can be altered in response to the body's level of hydration and hormonal cues such antidiuretic hormone (vasopressin). Both the water balance and urine concentration are regulated by this mechanism.
f) Vasopressin - P) Urine formation
Antidiuretic hormone (ADH), also known as vasopressin, is a hormone produced by the hypothalamus and released by the pituitary. By adjusting the collecting ducts' permeability in reaction to the body's level of hydration, it manages the water balance. Vasopressin reduces urine production and aids in the formation of concentrated urine when it is released by increasing the reabsorption of water from the filtrate.